Class 8
Biology
(Solutions)

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1. Crop Production and Management

Solutions

LEVEL-I

1. Correct option: (A) Summer

Vegetables are frequently readily accessible in a large range during the summer. Numerous veggies may grow well in the warm climate and longer days. During the summer, there is typically an abundance of fresh, colorful vegetables at grocery shops and farmers' markets. In addition to the large selection of veggies that are offered in the summer, this is also the season when local food is at its most flavorful and fresh. Because of their nutritious value and delectable flavor, many people choose to include these seasonal vegetables in their diets.

Vegetables like tomatoes, cucumbers, bell peppers, zucchini, maize, green beans, eggplant, and a variety of leafy greens like lettuce and spinach are a few examples of those that are frequently found throughout the summer.

2. Correct option: (B) June to September.

In India, the monsoon season is known as kharif and normally lasts from June to September. Farmers plant crops at this time of year because they need the water for the crops' growth and development.

The monsoon rains' arrival provides the moisture these crops need to flourish. Rice, maize, millet, sorghum, cotton, sugarcane, and numerous vegetables are among the crops grown during the kharif season.

Crops during the Kharif season are typically sown from the start of the monsoon season to September. The collected crops can subsequently be consumed or processed further.

3. Correct option: (A) Rice

In light of its extensive global cultivation and consumption, rice has the distinction of being the world's most consumed grain. For billions of people, it is a staple dish, particularly in Asian nations where it is the foundation of the local cuisine.

Rice farming is adaptable and may be done in a variety of climates, from dry ground to flooded paddy fields.

In addition to giving farmers work opportunities and sustaining rural lifestyles, rice farming makes a considerable contribution to agricultural economies. Because of its high nutritional value, low cost, and extended shelf life, rice is an essential component of the world's food supply and is used to satisfy the dietary demands of many different people.

4. Correct option: (B) Seed drill

A seed drill is a piece of agricultural equipment made particularly for planting seeds in the ground. It is utilized for tractor-assisted sowing and makes sure that seeds are coated in soil after planting. The way the seed drill works is by making channels or furrows in the earth, and then carefully planting the seeds at the precise depth and spacing.

This exact seed-planting technique encourages consistent germination and plant development, which boosts agricultural yields. In contrast to conventional manual techniques, saves time and labor while preventing birds from eating the seeds. The seed drill also facilitates appropriate seed-to-soil contact for efficient germination, optimizes seed dispersion, and reduces seed loss.

5. Correct option: (C) Srivari

The Srivari kind of paddy is the one that needs a bigger space between plants than the other varieties. This type is renowned for its robust growth and wide-ranging canopy, which need greater room between plants for the best growth.

Srivari paddy plants may obtain enough sunshine, air circulation, and nutrients by having enough space between them, which results in greater yield and healthier plants. To satisfy its growth requirements, farmers that cultivate Srivari paddy should maintain a greater space between plants.

6. Correct option: (A) Tikka disease

Tikka disease is a typical disease that affects groundnuts. The fungus Cercospora arachidicola is what causes tikka illness. Small, dark brown to black spots with a concentric ring pattern appear on the groundnut plant's leaves and pods as a result of this condition. The spots may get larger and combine as the illness spreads, which can cause defoliation, yellowing of the leaves and pods, and decreased production.

Tikka disease in groundnut agriculture may be controlled and its effects reduced with the use of effective management techniques such crop rotation, seed treatment, and foliar fungicide use.

7. Correct option: (D) Irrigation

Irrigation is the systematic, planned application of water to plants in order to promote their optimum development. In order to maintain soil moisture levels, it entails the controlled administration of water to crops at predetermined intervals. Since it guarantees that plants receive enough water for photosynthesis, nutrient absorption, and general growth, adequate irrigation is essential for crop health. The kind of crop, soil, weather, and development stage are just a few examples of the variables that affect when and how often irrigation should be applied.

Different irrigation techniques are used to effectively distribute water to crops, minimising water loss via evaporation and guaranteeing targeted delivery to plant roots. These techniques include sprinkler systems and drip irrigation systems. These techniques support sustainable agricultural practises and water resource conservation.

8. Correct option: (A) Dried

Grain must be adequately dried in order to lower its moisture content before being stored. Harvested grains frequently have significant moisture content, which, if left untreated, can cause deterioration and the development of microbes.

Using the sun to dry grains is a popular and efficient way to get rid of extra moisture. The grains are spread out in thin layers and exposed to sunshine to dry by evaporating the moisture. The quality and integrity of the grains are protected during storage by adequate drying, which inhibits the growth of bacteria, fungus, and insect pests that flourish in damp settings.

The acceptability of grains for food and retaining their viability for germination if used as seeds is ensured by well dried grains, which also have a lower risk of mould development, mycotoxin contamination, and insect infestation. Additionally, before storage, grains can be dried to increase their shelf life and maintain their nutritious content.

9. Correct option: (A) Broadcasting.

Broadcasting is the method of seed dispersal by sprinkling or scattering seeds evenly over a wide area. In agriculture, broadcasting is commonly used for sowing seeds in fields, especially for crops like grass, grains, and certain vegetables. The seeds are dispersed by hand or by using seeders, ensuring they are spread uniformly across the soil surface.

This method allows for a relatively even distribution of seeds, promoting uniform plant growth and reducing competition among individual plants for resources like water, sunlight,

and nutrients. Options B) Throwing and C) Flying do not accurately describe the method of seed dispersal, and option D) None of these is not the correct answer in this context.

10. Correct option: (C) Cold storage

Agricultural products may often be kept for a long time by being placed in cold storage. It entails keeping perishable goods at low temperatures, including various cereals, dairy products, and produce. The purpose of cold storage facilities is to maintain particular temperature and humidity levels that inhibit the growth of bacteria and enzymes that cause rotting and degradation. By maintaining low temperatures, typically ranging from 0 to 10 degrees Celsius (32^0 to 50^0 Fahrenheit), cold storage inhibits the activity of spoilage-causing agents and extends the freshness of the stored products.

Agricultural goods can stay fresh, maintain their nutritional content, and have a longer shelf life when they are kept in cold storage since the rate of deterioration is greatly lowered. Options A) Bamboo bin and B) Metallic bin are not specifically associated with preserving agricultural products for longer periods, and option D) None of these is not the correct answer in this context.

11. Correct option: (B) Sank seeds

Seeds that sink in water generally have a higher likelihood of germinating well due to several key factors. Firstly, sinking seeds indicate higher seed viability, suggesting that they are alive and capable of germination. Secondly, sank seeds often have intact seed coats, providing protection and allowing water to penetrate for germination. Additionally, these seeds typically possess well-developed embryos, containing essential structures for growth. Sank seeds also tend to have adequate internal reserves of nutrients and energy, providing nourishment during early stages of germination.

Moreover, sinking seeds are less likely to have encountered damage or decay, reducing the risk of unsuccessful germination. While the sinking or floating behavior is an approximate indicator, other factors such as seed age, genetics, and specific environmental requirements also influence germination success.

12. Correct option: (A) Rabi

Wheat is primarily cultivated during the Rabi season due to its favorable climatic conditions and the crop's growth requirements. First of all, because wheat is a cool-season crop that prefers moderate temperatures for growth, the Rabi season's milder weather is suitable.

Second, the availability of water is very important. During the Rabi season, the soil moisture left over from the previous monsoon season is often sufficient, decreasing the need for irrigation.

Thirdly, the Rabi season's shorter days and longer nights give wheat the proper photoperiod for its reproductive development stage. In addition, the number of pests and illnesses that generally harm wheat decreases during the Rabi season, lowering the possibility of crop loss. Finally, the Rabi season harvest coincides with the arrival of spring, ensuring an adequate supply of wheat for consumption and distribution.

13. Correct option: (A) Rabi

The term "Rabi crops" refers to plants that are only cultivated during the winter. Crops during the rabi season are planted in the months of October to December and harvested in the following spring. These crops include, among others, wheat, barley, mustard, peas, gramme, and linseed. Since these crops are grown in the winter and harvested in the spring, the word "Rabi" is derived from the Arabic word for "spring." The milder temperatures, prolonged photoperiods, and lingering monsoon-season soil moisture are beneficial to rabi crops. They are essential in maintaining the availability of food and giving farmers a reliable source of income during the winter.

14. Correct option: (B) Kharif

Kharif crops are those that are raised during the rainy season. Crops known as kharif are planted at the start of the monsoon season in June or July, and they are harvested from September to October. These crops include, among others, groundnuts, cotton, rice, maize, millet, and sorghum. Given that they need more water for growth and development, kharif crops thrive in the hot, muggy weather of the rainy season. They usually have a lengthier growth cycle compared to Rabi crops and mainly rely on the monsoon rains for irrigation. In areas with a distinct monsoon pattern, kharif crops are essential for agriculture because they make a substantial contribution to total food supply and the rural economy.

15. Correct option: (D) All the above

Urea, DAP (diammonium phosphate), and potash are a few typical chemical fertilizers used in agriculture. The fertilizer urea, which is based on nitrogen, gives plants the nitrogen they need for leaf formation and general growth. The root growth, blooming, and fruiting of plants are supported by the phosphorus-based fertilizer known as DAP. The term "potash," on the other hand, refers to fertilizers that contain potassium, a mineral that is crucial for plant health, disease resistance, and a number of physiological functions. These chemical fertilizers are essential in refilling the soil's nutrient content, enhancing plant nutrition, and increasing crop yields. To reduce any possible negative effects on the environment, it is crucial to utilize them wisely and in the right amounts.

16. Correct option: (C) V

V-shaped ridges are indeed formed in certain farming practices when ploughing is done. This technique is known as "ridge and furrow" or "ridge tillage." It involves creating raised V-shaped ridges of soil with furrows in between. The purpose of this method is to improve water management and drainage in areas with high rainfall or poor soil drainage. The V-shaped ridges help divert water away from the planting area, preventing waterlogging and promoting better root aeration. The furrows between the ridges collect excess water and serve as channels for irrigation or drainage. This technique is commonly used in certain crops such as potatoes, where it helps in improved root development and better access to sunlight and nutrients.

17. Correct option: (B) 2300 BC

The Harappan civilization, thriving around 2300 BC in the Indus River valley, displayed remarkable advancements in crop management and cultivation. Agriculture played a crucial role in their economy, and they cultivated a variety of crops. The Harappans employed sophisticated techniques for irrigation, utilizing an extensive network of canals and wells to ensure a steady water supply for their fields.

The irrigation systems supported year-round cultivation, enabling multiple cropping seasons and maximizing agricultural productivity. They practiced systematic land preparation, including ploughing and leveling, to create fertile fields. The Harappans also employed innovative strategies for water management, such as the construction of reservoirs and the implementation of water storage and distribution systems. Evidence suggests that they practiced crop rotation to maintain soil fertility and prevent nutrient depletion.

Additionally, the Harappans may have utilized organic manure and composting methods for soil enrichment. Their agricultural practices contributed to the sustenance and prosperity of the civilization, ensuring a stable food supply for its population.

18. Correct option: (C)

19. Correct option: (A) 63^0

In pasteurization, milk is heated at a temperature of approximately 63^0 Celsius (145 degrees Fahrenheit). This specific temperature is chosen as it effectively kills harmful bacteria and

pathogens present in the milk, reducing the risk of foodborne illnesses. The heat treatment destroys most of the pathogenic microorganisms while still preserving the nutritional properties and sensory characteristics of the milk.

By subjecting milk to this controlled heating process, pasteurization ensures that consumers can enjoy a safer product without compromising the taste and quality. Pasteurized milk is widely consumed and trusted as a safe source of nutrition.

20. Correct option: (D) Chilka

The Chilka buffalo is the traditional breed of buffalo used for cattle in Odisha. In particular, this breed has been suited to the marshes and marshy terrain of the Chilka Lake region of Odisha. Chilka buffaloes are prized for their prowess in this environment, which makes them suitable for grazing and draught uses. They are also well recognised for producing milk, which helps the local dairy sector. The Chilka buffalo is significant culturally and economically in Odisha, where it is crucial to local populations' lives and agricultural practises.

21. Correct option: (B) Eggs

Layers refer to a specific type of poultry breed that is primarily raised for egg production. These birds are selectively bred to lay a large number of eggs throughout their productive life. They are known for their high egg-laying capacity and can produce a significant quantity of eggs on a regular basis. Layer chickens are typically housed in specialized facilities with proper lighting, temperature control, and nutrition to ensure optimal egg production. The eggs produced by layers are a valuable source of protein and are widely consumed as a nutritious food item.

22. Correct option: (D) January to April

The months of January to April generally see higher egg prices. During this period, there are various factors that contribute to the increased prices, such as increased demand due to seasonal festivities like Easter and the onset of spring when people tend to consume more eggs. Additionally, during these months, there may be fluctuations in the supply of eggs due to factors like weather conditions, changes in feed availability, and production cycles. These factors collectively contribute to the higher prices of eggs during the months of January to April.

23. Correct option: (A) Aseel

Aseel, also known as Asil or Asli, is a breed of chicken that is well-known and valued for its fighting abilities. Aseel chickens have a long history in India and are highly regarded for their strength, endurance, and aggressive nature. They are bred specifically for cockfighting, a traditional sport in which two Aseel roosters are pitted against each other in a fight. Aseel chickens are characterized by their muscular build, strong beaks, and distinctive appearance. They are considered a prized breed among enthusiasts of cockfighting and are highly sought after for their fighting prowess.

24. Correct option: (A) 40kg

Emus are the second-largest living birds by height and are native to Australia. These flightless birds have a large and sturdy build, and adult Emus typically weigh around 45 to 60 kg (99 to 132 pounds). However, there can be variations in the weight of individual birds based on factors such as age, gender, and overall health. The weight of an Emu bird can also be influenced by environmental factors and the availability of food resources.

25. Correct option: (D) 1

In a honey bee colony, there is typically only one queen bee. The queen bee is the sole reproductive female in the colony and plays a crucial role in the survival and functioning of the colony. She is larger in size and specially developed for egg-laying. Her primary responsibility is to lay fertilized eggs, which will develop into worker bees and new queens.

The queen bee releases pheromones that maintain harmony and coordination within the colony. The presence of multiple queen bees in a single colony can lead to conflicts and disruptions in the social hierarchy. Therefore, in a healthy and stable honey bee colony, there is generally one queen bee.

26. Correct option: (B) Agriculture

Agriculture encompasses various activities related to the cultivation of crops and the rearing of animals. Animal husbandry specifically focuses on the management and care of domesticated animals for agricultural purposes. It involves practices such as breeding, feeding, housing, and healthcare of livestock, including cattle, poultry, sheep, goats, and pigs. Animal husbandry plays a vital role in agriculture as it provides a sustainable source of food, fiber, and other products derived from animals. It contributes to the overall agricultural economy and helps meet the growing demand for animal-based products such as meat, milk, eggs, and wool. Additionally, animal husbandry practices are crucial for maintaining the health and productivity of livestock, ensuring optimal production and maximizing agricultural outputs.

27. Correct option: (D) Nagori

The Nagori is a traditional variety of cow found in certain regions of India. The regions of Rajasthan, India, known as Bikaner, Jodhpur, and Nagaur, are home to the Nagori breed of cattle. With a long, thin face like a horse's, these animals are renowned for their upright, nimble character. Due of their close closeness to the Hariana and Kankrej breeds, it is probable that these breeds have had an impact on them.

The bodies of nagori cattle are excellent, massive, and deep, with straight backs and well-developed quarters. They are known for their trotting draught prowess and are frequently utilised in light iron-wheeled carts for swift transportation in the Rajputana region of Rajasthan.

These cows produce 603 kg of milk on average every lactation, with a milk fat level of 5.8% on average. Nagori cattle are valued for their distinctive qualities and cultural significance in the local populations. They also support local agricultural practises.

28. Correct option: (D) Both A and B

D) Both A) Jersey and B) Holstein.

The Jersey breed is known for its milk production capabilities, yielding an average of 5,000-8,000 kg of milk per lactation. A Jersey cow can produce around 20 liters of milk per day, making it a high-yielding breed. Crossbred Jersey cows also have good milk production, typically giving 8-10 liters per day. Jerseys are well-suited to hot and humid climates, making them suitable for the climatic conditions of India.

On the other hand, the Holstein breed is considered one of the best dairy breeds among exotic cattle in terms of milk yield. They have an average milk yield of 7,200-9,000 kg per lactation and can provide around 25 liters of milk per day. Crossbred Holstein cows can produce 10-15 liters of milk per day. Holsteins are large, capacious animals and are adaptable to coastal and delta areas.

29. Correct option: (A) 63^{0}C for 30 mnts

During pasteurization, milk is heated to a temperature of approximately 63°C (145°F) and maintained at that temperature for about 30 minutes. This specific temperature and duration are carefully chosen to effectively destroy harmful bacteria and pathogens present in the milk, ensuring its safety for consumption. The heat treatment kills potential pathogens such as Salmonella, Escherichia coli (E. coli), and Listeria, reducing the risk of foodborne diseases.

The pasteurization process also helps extend the shelf life of milk by inhibiting the growth of spoilage-causing microorganisms. By subjecting milk to this controlled heat treatment,

pasteurization strikes a balance between eliminating harmful bacteria while retaining the nutritional qualities and sensory attributes of the milk.

30. Correct option: (A) Food grains

From the middle of the 1960s onward, India saw a period of agricultural transformation known as the "Green Revolution." It required implementing cutting-edge agricultural techniques and tools, including the use of High Yielding Variety (HYV) seeds, better irrigation techniques, herbicides, and fertilisers. M.S. Swaminathan, known as the "Father of the Green Revolution" in India, was a key figure in advancing these ideas.

In areas like Punjab, Haryana, and Uttar Pradesh, the Green Revolution led to a huge rise in food grain output, particularly for grains like wheat. Modern agricultural techniques and the introduction of HYV seeds industrialised Indian agriculture, increasing output and alleviating food shortages.

31. Correct option: (A) 3rd

India is the third-largest egg producer in the world. The poultry business in the nation has expanded significantly, which has led to a huge rise in egg output. The Indian poultry industry has been effective in supplying eggs to fulfil the expanding local and international demand. India has become a significant producer of eggs, contributing significantly to global egg production because to improvements in chicken farming techniques, enhanced genetics, and effective production techniques.

32. Correct option: (C) 6th

India is the world's sixth-largest producer of chicken. With rising demand for chicken meat as a source of protein, India's poultry sector has seen significant expansion. To fulfil the expanding demands of customers, the nation has achieved considerable advancements in breed improvement, production methods, and commercial poultry farming practises. In addition to meeting domestic and international demand for chicken meat, India's poultry industry is essential for creating jobs and boosting the nation's economy.

33. Correct option: (C) Meat and eggs

Breeds of chicken that are often raised for both meat and eggs include Rhode Island Red and White Plymouth. These breeds are well-known for having traits that make them good for both producing meat and eggs. They may lay a sizable amount of eggs and have high-quality meat.

Due to its capacity to provide a sizable number of both meat and eggs, the White Plymouth Rock breed, also known as White Rocks, is a preferred option for commercial poultry production. On the other hand, Rhode Island Reds are renowned for having good meat yields along with exceptional egg-laying ability. Overall, these breeds are favoured by farmers who seek to maximise the amount of meat and eggs their chickens produce.

34. Correct option: (A) 21 days

Prior to hatching, hens normally give their eggs a 21-day incubation period. The hen supplies the essential warmth and humidity during this time so that the eggs can effectively grow and hatch. The majority of chicken breeds naturally incubate their eggs for 21 days. In order to guarantee appropriate embryo growth during this period, it is crucial to maintain the eggs at a constant temperature. The eggs will hatch and the chicks will break out of their shells after 21 days. The continuance of the chicken species depends on this period, which is a crucial component of the reproductive cycle.

LEVEL-II

1. Correct option: (C) Management of plants and animals

Animal and plant management are within the purview of agricultural science. It includes a diverse variety of academic fields and professional techniques designed to raise agricultural production and sustainability. Understanding and using knowledge of plant biology, crop production, soil science, animal husbandry, and genetics are required in this discipline.

The goal of agricultural science and practise is to improve the care and management of animals in agricultural systems as well as the development, health, and productivity of plants, such as crops. The science of agriculture aims to enhance food production, environmental preservation, and the general wellbeing of agricultural systems and societies by combining the management of both plants and animals.

2. Correct option: (A) Mentha

The herb mentha, often known as mint, is not generally categorised as a weed. The fragrant leaves of mint plants are frequently grown for use in food preparation and medicine. Mint is often not regarded as a weed species, despite the fact that it has the potential to spread and become invasive in some situations.

On the other hand, weed species including amaranthus, chenopodium, and wild oat are well known. A grassy plant called wild oat (Avena fatua) can lower productivity by competing with crops. A group of plants known as Chenopodium include a number of common weed species, including lamb's quarters (Chenopodium album). The plant genus Amaranthus, sometimes referred to as pigweed, has a number of troublesome species in agricultural fields.

3. Correct option: (C) By practicing crop rotation

Crop rotation can be used to restore soil fertility. Crop rotation is the methodical, deliberate alternating of several crops in a set order throughout time. Every crop has varied nutritional needs and engages with the soil in various ways. By changing up the crops that are grown, soil fertility is increased and nutrient depletion is prevented. Different crops also improve soil health by retaining or supplying certain nutrients, disrupting the life cycles of pests and diseases, lowering weed pressure, and improving soil structure. Crop rotation is a practical and efficient way to manage and replace soil fertility in agricultural systems.

4. Correct option: (B) Aluminium sulphate

Aluminium sulphate is not regarded as a chemical fertiliser. Instead of serving as a source of nutrients for plant development, it is largely utilised to modify soil and treat water. In several industrial and agricultural applications, aluminium sulphate is used to reduce the pH of water and to modify the pH of soil in alkaline soils. On the other hand, chemical fertilisers include urea, potash, and NPK.

Fertilisers that are based on nitrogen include urea, potash, and NPK, which is a complex fertiliser made up of nitrogen, phosphorus, and potassium. These chemical fertilisers are especially designed to provide vital nutrients that support plant growth, development, and productivity.

5. Correct option: (A) Cu, Zn, B

Micronutrients are necessary substances that plants must have in relatively little amounts, yet they are crucial to many physiological functions. Micronutrients include elements such as boron (B), copper (Cu), and zinc (Zn). These micronutrients are essential for the development, growth, and general health of plants. Photosynthesis and enzyme activation both include copper. Protein synthesis, hormone control, and enzyme activity all depend on zinc. The synthesis of cell walls, pollen germination, and nutrient absorption are all influenced by boron. These micronutrients are crucial for healthy plant development and are frequently added to the soil or fertiliser for plants when shortages are noticed.

6. Correct option: (C)

When harvesting and threshing crops like wheat, corn and soybeans, a combine, often referred to as a combine harvester, is employed. The crop harvesting process is streamlined by combining the duties of harvesting and threshing into a single operation. To collect mature crops like wheat stalks or maize plants, the combine utilises a cutting mechanism. The threshing process separates the grain (such as wheat or maize kernels) from the remaining plant matter (stalks, husks, etc.) as the harvested crop enters the machine.

In the combine, the separated grain is gathered in a storage tank and made ready for processing or transportation. The remainder of the plant is ejected or thrown away. By drastically lowering labour, combines have revolutionised agriculture.

7. Correct option: (D) Sprinkler system

Rotating nozzles are utilised in the Sprinkler System while discussing irrigation techniques. Sprinkler systems are intended to disperse water across fields similarly to rainfall. By spraying water in a revolving pattern, the spinning nozzles are essential in providing uniform water coverage.

The sprinkler heads revolve to ensure that water is uniformly distributed around the field, boosting effective water use and minimising water waste. This technique is frequently used in agriculture for extensive irrigation since it enables efficient watering of a sizable region while upholding water conservation and maximising crop water needs.

8. Correct option: (A) Traditional method of irrigation

Moats are tiny holes or pits that are excavated to plant seeds or seedlings; this has less to do with irrigation and more to do with seed sowing or planting methods.

Instead of irrigation, dhekli is a technique used to flatten or crush soil clods following ploughing, which is a part of land preparation.

Rahat is a method of soil preparation that also entails turning over and loosening the soil with a tool like a plough.

Moat, Dhekli, and Rahat do not fall under the conventional technique of irrigation if we just take option A) into account.

9. Correct option: (B) Beans

Starchy grains that are predominantly utilised as food sources are produced by plants known as cereal crops. These grains are significant staple crops globally and are frequently high in carbs. Wheat, millet, and barley are a few examples of cereal crops.

Conversely, beans are a member of the family of legumes. Various plants that produce pods with seeds within are referred to be legumes. Although they are valued and very nutritious, beans are not considered a cereal crop. Beans, on the other hand, belong to the legume family and are categorized as pulses or legume crops. They have a greater protein content than cereals and are renowned for their capacity to fix nitrogen in the soil, enhancing soil fertility.

10. Correct option: (C) Dried neem leaves

Since dried neem leaves naturally resist insects, they are utilised as grain storage at home. Neem leaves contain substances like azadirachtin, which are powerful insect repellents. Neem leaves are completely dried to improve their preservation qualities before being used to store grains. After that, they are positioned or dispersed among the grains that have been kept to form an insect and pest-resistant barrier.

The dried neem leaves' strong perfume serves as a natural insect repellent, keeping pests away from the grains. Using dried neem leaves in this manner is a traditional strategy to safeguard grains that are kept in storage, assuring their long-term preservation and averting any harm from pests or insects.

11. Correct option: (C) Cattle dung and crop residue

Farmyard manure (FYM), sometimes referred to as organic manure or FYM, is a form of natural fertiliser and soil amendment that is produced by the decomposition of a combination of agricultural debris and animal dung. A high amount of organic matter and nutrients is available in cattle dung, which is made up of the faeces and urine of cattle.

The organic component of the manure also benefits from agricultural waste, such as residual plant matter from harvested crops. These materials breakdown naturally, such as by composting, generating a nutrient-rich mixture that may be added to soil to increase its fertility, structure, and general health.

12. Correct option: (B) Roots of leguminous plants

The majority of rhizobium bacteria are found in nodules that form on the roots of leguminous plants. Rhizobium bacteria are recognised for their capacity to fix atmospheric nitrogen. The Rhizobium bacteria and legume plants have a special symbiotic connection. The bacteria populate the root nodules and transform air nitrogen into a form the plant can use.

Leguminous plants can obtain a crucial ingredient for their growth and development through a process known as nitrogen fixation. Leguminous plants may grow in nutrient-poor soils thanks to Rhizobium, which is found in the root nodules. Rhizobium also plays a critical role in enhancing soil fertility by converting atmospheric nitrogen into a form that can be used by plants.

13. Correct option: (D) Ganga 5

Among the available possibilities, Ganga 5 is the hybrid variety of maize. Different parent lines are crossed to create hybrid types in order to produce offspring with the desired qualities. A hybrid maize variety with certain properties such a higher yield potential, improved disease resistance, or other beneficial attributes has been carefully cultivated, and this trait is what is meant by the name Ganga 5. Due to its enhanced characteristics, this specific hybrid variety has become well-known and well-liked among farmers and is frequently used in agricultural practises.

14. Correct option: (A) Plough, hoe cultivaor

The plough, hoe, and cultivator are the primary implements used to break the soil into the correct grain size and provide a greater yield before sowing the seeds.

A instrument used to turn over and loosen the soil is the plough. It assists in loosening up compacted soil, burying agricultural waste, and improving soil drainage and aeration.

A portable device called a hoe has a flat, sharpened blade. It is used to break up soil clumps, get rid of weeds, and provide a smooth, level seedbed.

The cultivator is a device with several blades or tines. It is used to cultivate the soil only superficially, break up clumps, and incorporate organic materials or fertilisers into the soil's topmost layers.

15. Correct option: (D) Seeds placed deep into soil

Seeds are typically planted at a suitable depth, which varies depending on the type of seed and plant. Planting seeds at an appropriate depth is crucial for their successful germination and growth. When seeds are buried too deep in the soil, they may struggle to reach the surface and access sunlight, hindering their ability to undergo photosynthesis.

Additionally, planting seeds too deep can limit their access to moisture, impeding their ability to absorb the necessary water for germination. Furthermore, an inadequate exchange of gases between the seed and its surroundings may occur, depriving the seed of the oxygen it needs for metabolic processes. Therefore, it is important to follow the recommended planting depth for each type of seed to provide them with the best conditions for sprouting, establishing roots, and developing into healthy plants.

16. Correct option: (C) Drip system

The drip irrigation system is not a common type of irrigation. It is a contemporary irrigation technology that uses a system of tubes or emitters to precisely and carefully apply water to the plant's root zone. Drip irrigation is a prominent option in contemporary agricultural practises because of its reputation for water efficiency and precise delivery.

On the other hand, conventional irrigation techniques include the chain pump, pulley system, and lever system. These techniques, which have been used for millennia, employ mechanical or human means to move water for irrigation of crops.

17. Correct option: (D) Rhizome

A rhizome, often known as an underground modified stem, is an example of ginger. There are nodes and internodes on rhizomes, which are horizontal subterranean stems. The rhizome of ginger acts as a nutrient storage organ and promotes vegetative growth. The ginger plant may extend and expand as a result of the new branches and roots it generates. Ginger's rhizome is an important and well-known example of an underground modified stem since it is picked for both culinary and medicinal purposes.

18. Correct option: (D) None of the given options

A) Plants benefit from the growth of living organisms like bacteria and earthworms. It's accurate to say this. When it comes to the health of the soil and the cycling of nutrients, earthworms and microbes are key players.

B) The live component is supported by non-living elements. This assertion is true. Living things, such as plants and microbes, depend on non-living elements like soil, air, water, and minerals for their physical and chemical sustenance.

C) The seed drill saves labour and time. This assertion is true. In contrast to conventional hand sowing techniques, a seed drill is a mechanical tool used for accurate and effective seed planting.

Since all the given statements are correct, the correct answer is D) None of the given options.

19. Correct option: (C) Neem leaves

Neem leaves have been used for centuries as a natural pesticide. They include substances that serve as antifeedants, insect repellents, and growth and reproduction disruptors. Crushed neem leaves, neem oil extracts, or formulations containing neem can all be used to protect crops from insects and pests. Neem leaves also have herbicidal qualities that inhibit the growth of weeds in agricultural regions.

Neem leaves are well known for their strong pesticidal powers and are frequently employed by farmers to safeguard their crops against weeds and insect pests. Other leaves, such as basil leaves, peepal leaves, and mango leaves, may have diverse useful features.

20. Correct option: (A)

The International Rice Research Institute (IRRI), which is situated in the Philippines, brought a variety of rice known as the IR variety to India. The IR rice variety is well known for its qualities of disease and insect resistance, which have made it an important contribution to rice agriculture in several nations, including India. The introduction of the IR variety has improved crop resilience, increased rice yield, and aided with efforts to ensure food security.

This sentence makes it clear that the IR variety is only related to rice and that it was brought to India from the Philippines. Rice production has benefited greatly from the IR variety, which has given farmers access to enhanced crop types that are resistant to a wide range of diseases and pests. This rice variety was introduced to help farmers in India and other nations overcome agricultural obstacles and improve the overall productivity and calibre of rice crops.

21. Correct option: (C) Sharbati, Sonara

It is well known that Dr. M.S. Swaminathan made significant advances to wheat farming during India's Green Revolution. His creation and promotion of high-yielding dwarf wheat cultivars like "Kalyan Sona" and "Sonalika" revolutionised the industry, boosting yields and solving issues of food security. His work had a tremendous influence on Indian agriculture, enhancing wheat farming and assisting in ensuring that the populace had access to food.

Even though "Sharbati Sonara" isn't particularly linked to Dr. M.S. Swaminathan, it is a well-liked wheat variety in some parts of India that is renowned for having a high protein content. However, it's critical to recognise how Dr. Swaminathan's introduction of high-yielding dwarf cultivars has had a larger influence on wheat farming in India.

22. Correct option: (B) It has a balance of all plant nutrients

Compost or farmyard manure are examples of organic manures that come from natural sources like plant remnants, animal dung, or organic wastes. Although organic manure is good for the health and fertility of the soil, it can not always contain a balance of all the plant nutrients.

Nitrogen, phosphorus, and potassium are among the many elements that organic manure normally includes, although the amount of each component varies depending on the source and makeup of the manure. Comparatively to synthetic fertilisers, which are especially designed to offer a balanced supply of vital plant nutrients, organic manure often has a lower nutritional concentration.

23. Correct option: (B) Leguminous plant can fix nitrogen

Unique to legume plants, rhizobia, which fix nitrogen, create a symbiotic connection with legume plants. These microorganisms are found in nodules in the roots of leguminous plants, where they transform atmospheric nitrogen into a form the plants can use for growth. Leguminous crops may naturally replenish the soil with nitrogen through a process known as nitrogen fixation, which is one of its major advantages.

Option a) Chain pumps are not the standard irrigation technique. Traditional irrigation systems use methods such employing canals, wells, or water channels.

Option C) Soil nutrients are not always depleted when various crops are grown in different seasons on the same field. In reality, by limiting the excessive depletion of certain minerals and lowering the danger of pests and diseases, crop rotation and diverse cropping systems can assist preserve soil fertility.

24. Correct option: (C)

The process of planting two or more crops at once on the same plot of land is known as intercropping. It entails planting various crops next to one another in a planned sequence or arrangement. The crops used for intercropping may have advantageous interactions, complimentary growth patterns, or nutritional requirements. By diversifying crops and encouraging optimal use of sunshine, water, and nutrients, intercropping helps maximise land production, optimise resource usage, and lower the chance of crop failure.

25. Correct option: (A) Give rise to sexually sterile plants

The 'Terminator Seed Technology,' often referred to as Genetic Use Restriction Technology (GURT), is a contentious method of genetically altering plants to generate sterile or incapable of germination seeds. This suggests that option A, sexually sterile plants, should grow from the seeds generated by the method.

Farmers are compelled to buy new seeds for each planting season due to the terminator seed's ability to prohibit farmers from conserving and recycling seeds from previous crops. Farmers are extremely concerned about this element of terminator seeds since it limits their customary practise of seed storage and traps them in a cycle of dependency on seed providers.

26. Correct option: (D)

27. Correct option: (C) Only 1 and 4

Vitamin A: A lot of vitamin A may be found in fish liver oil. The immune system, promotion of healthy skin, and maintenance of excellent vision all depend on vitamin A.

Vitamin D: Vitamin D is also abundant in fish liver oil. Vitamin D is essential for boosting calcium and phosphorus absorption, supporting strong bones and teeth, and sustaining general bone health.

28. Correct option: (A)1,2 & 3

Introduction: To improve a crop population's qualities, additional genetic material or varieties are bred into the population. This may entail bringing in fresh plant kinds from other areas or adding genetic material using strategies like genetic engineering.

Hybridization: The process of breeding wherein two genetically dissimilar plant species are crossed to produce a hybrid progeny. Hybridization can produce plants with desired features like higher production, disease resistance, or enhanced quality.

Selection: Selecting and carefully breeding plants from a population of plants that have qualities you want. Plants that exhibit qualities like increased yield, better quality, or resistance to pests and diseases are chosen and allowed to breed, which over several generations improves the crop population.

Segregation: Refers to the separation of characteristics rather than active improvement, is not required for raising crop quality.

The techniques required to increase crop quality include introduction (introducing new varieties), hybridization (crossing various varieties), and selection (selecting and propagating favourable features). As a result, the right response is A) 1, 2 & 3 (Introduction, Hybridization, and Selection).

29. Correct option: (C) Only 1, 2 and 3

Paddy (Rice): Growing paddy or rice frequently uses the transplanting method. With this technique, rice seedlings are first raised in a separate nursery field and then, once they reach a specific development stage, moved to the main paddy field. In addition to ensuring ideal growth conditions for the rice crop, this enables improved control over plant spacing.

Tomato: Growing tomato plants is frequently accomplished by transplantation. First, tomato seeds are sown inside or in a greenhouse in seed trays or pots. The seedlings are moved into the outdoor garden or field once they have grown to an appropriate size and strength. This technique, especially in areas with shorter growing seasons, enables early germination and gives the plants a head start.

Chilli (Chili peppers): Similar to tomato plants, chilli peppers are frequently cultivated using the transplanting technique. The seedlings can develop and flourish in the best conditions since the seeds are originally planted in controlled surroundings like seed trays or pots. The chilli pepper seedlings are moved into the actual growing area, such as a garden or field, once they reach a particular size.

Millet: Millet crops are typically not grown using the transplantation method. Instead, millet seeds are usually directly sown into the ground or field where they will grow.

30. Correct option: (C) a – ii; b – i; c – iii

a) Sowing: The act of dispersing seeds for growth is referred to as sowing. The first stage of planting crops is this. Option ii correctly identifies this agricultural practise as "sowing," which is the act of dispersing seeds on the earth.

b) Irrigation: Water is supplied to crops in the fields during irrigation. It is crucial for giving plants the moisture they need for development and growth. This practise is adequately described by option i, which explains that irrigation is the process of providing water to crops in the fields.

c) Weeding: The act of eliminating undesirable plants from a field is referred to as weeding. It entails getting rid of undesired plants or limiting their development so they don't compete with cultivated crops for resources like water, nutrients, and sunshine. Weeding is correctly described as the process of eliminating undesirable plants from a field in Option iii.

2. Micro-organisms

Solutions

LEVEL – I

1. Correct option: (A) Curd

Curd, a fermented type of milk, is produced by the action of specific Lactobacillus genus rod-shaped bacteria. These lactic acid bacteria are gram-positive, non-spore-forming bacteria. Lactobacillus bacteria, which metabolize the carbohydrates in milk, produce lactic acid, which drives the conversion of milk to curd.

Lactobacillus is a well-known probiotic component present in foods such as yoghurt and is associated with several health benefits. It assists in the restoration of homeostasis and is especially effective in situations of IBS. The Lactobacillus genus has several distinct species, and curd may include Lactobacillus acidophilus, Lactobacillus lactis, and Lactobacillus lactis cremoris, among others.

2. Correct option: (C) Yeast

Due to its critical function in the fermentation process, yeast, namely Saccharomyces cerevisiae, is added to dough for bread making. When yeast is added to the dough, it metabolizes the carbohydrates present and ferments them into carbon dioxide gas and alcohol. This gas becomes trapped within the dough, allowing it to rise and form air pockets, giving bread its distinctive light and fluffy feel.

Yeast also helps to the flavor development of bread by creating a variety of organic chemicals, including ethanol, and adding a pleasant scent. Yeast is an essential element in bread-making due to its capacity to leaven dough and generate the right texture and flavor.

3. Correct option: (A) Carbon dioxide

Fermentation is a metabolic process performed by microorganisms such as yeast or bacteria that includes the breakdown of organic compounds such as sugars or carbohydrates in the absence of oxygen. This process breaks down complex organic chemicals into simpler compounds such as alcohol or acids, releasing energy in the form of ATP. Carbon dioxide (CO2) is a particularly noticeable byproduct of fermentation.

This gas is produced by microbes as a byproduct of the breakdown of sugars or carbohydrates and is emitted into the environment as waste. Carbon dioxide generation is widespread in fermentation processes such as alcoholic beverage manufacturing, bread manufacture, and beverage carbonation.

4. Correct option: (C) Alcohol

Yeast is a kind of microbe that plays an important part in fermentation processes. When yeast metabolizes carbohydrates, it goes through a sequence of biochemical events known as alcoholic fermentation. During this process, yeast transforms carbohydrates such as glucose or fructose into alcohol (particularly ethanol) and carbon dioxide. This conversion happens in the absence of oxygen and culminates in the creation of alcohol as the principal end product.

As a result, when carbohydrates are metabolized by yeast, they are transformed to alcohol. Option C) Alcohol is the proper response in this situation.

5. Correct option: (A) Antibiotics

Antibiotics are drugs that are used to treat bacterial illnesses by either killing or inhibiting the development of the bacteria. They accomplish this by specifically targeting certain systems or structures inside the bacterium. Antibiotics efficiently weaken or eradicate bacteria by interfering with critical activities such as cell wall formation or protein biosynthesis. This aids in the cure or treatment of bacterial infections by eradicating the

infection and alleviating related symptoms. Antibiotics are ineffective against viral infections and should only be taken when recommended by a healthcare practitioner to ensure proper and ethical use.
Painkillers (option B) help in managing pain associated with various conditions but do not directly cure the underlying disease or infection. Nutrients (option C) are essential for maintaining overall health and supporting the body's immune system, but they do not cure diseases on their own. Antigens (option D) are substances that stimulate the immune system to produce antibodies, but they are not used to directly cure diseases.

6. Correct option: (D)

7. Correct option: (C) Cow

The name "vaccine" is derived from the Latin phrase "Vaca," which translates to "cow." This relationship is based in the history of vaccination, which was pioneered by Edward Jenner. Jenner discovered that milkmaids who had cowpox were resistant to smallpox.
He later invented a way of inoculating humans against smallpox using material from cowpox lesions, which marked the beginning of vaccination. The name "vaccine" was developed to represent this relationship to cowpox and to recognize the essential role of cows in the creation of this medical achievement.

8. Correct option: (D) Nitrogen

Nitrogen makes up around 78% of the air we breathe. The most abundant gas in the Earth's atmosphere is nitrogen. Oxygen comes in second, accounting for around 21% of the air we breathe.
Nitrogen is a necessary component of life, playing an important part in biological processes and promoting plant development. It also participates in the nitrogen cycle, where it undergoes several changes before being used by organisms. Furthermore, nitrogen helps to maintain atmospheric stability by diluting other gases and inhibiting fast burning.
While other gases such as carbon dioxide, helium, and trace gases account for the remaining 1%, nitrogen dominates our atmosphere and is essential for ecosystem function and the survival of life on Earth.

9. Correct option: (B) Bacteria

Lemon citrus canker is caused by bacteria. Citrus canker is a plant disease caused by the bacterium Xanthomonas citri subsp. citri that damages citrus plants, notably lemons. citri. This bacterium affects citrus trees' leaves, stems, and fruit, causing elevated corky sores on the plant's surface. Wind, rain, and infected tools or equipment are all ways for the bacteria to spread.
Citrus canker can have a substantial economic impact since it affects the market value of afflicted fruits and may lead to tree defoliation and decline. Quarantine procedures, trimming of affected plant portions, and the use of copper-based sprays to suppress the bacteria are all effective management options.

10. Correct option: (B) Spoiled food

Food poisoning is caused mostly by eating damaged or contaminated food. When food is infected with hazardous bacteria, viruses, parasites, or their toxins, it causes food poisoning.
Pathogens can proliferate in food that has not been properly kept, handled, or cooked. Consumption of rotten or contaminated food can cause nausea, vomiting, diarrhea, stomach discomfort, and, in extreme cases, dehydration and organ damage.
To avoid foodborne infections, good food hygiene practices must be followed, including correct food storage, cooking, and handling. Unless infected with hazardous germs, fresh fruits, uncooked food, or poor air quality are unlikely to induce food poisoning.

11. Correct option: (A) 1928

Penicillin, the first widely used antibiotic, was discovered in 1928. It was accidentally discovered by Alexander Fleming, a Scottish bacteriologist, while he was studying bacteria in his laboratory at St. Mary's Hospital in London. Fleming noticed that a mold called Penicillium notatum had contaminated one of his petri dishes, and the bacteria surrounding the mold were being killed.

This observation led to the discovery of penicillin, which revolutionized the field of medicine by effectively treating bacterial infections. The discovery of penicillin laid the foundation for the development of numerous other antibiotics and significantly improved the treatment of infectious diseases.

12. Correct option: (C) Tetracycline

Tetracycline, an antibiotic that has been essential in the fight against bacterial diseases, was discovered by Dr. Y. Subba Rao, who is recognized for this discovery. Dr. Subba Rao initially discovered tetracycline in a soil sample he took in the late 1940s.

This all-purpose antibiotic stops bacterial growth and replication by preventing them from synthesizing proteins. Since its discovery, tetracycline has been widely used to treat a variety of bacterial illnesses, such as skin infections, urinary tract infections, and respiratory tract infections. And also used in medicine and agriculture.

Numerous lives have been saved because to Dr. Subba Rao's ground-breaking research that led to the discovery of tetracycline and set the path for the creation of additional critical antibiotics in the battle against infectious illnesses.

13. Correct option: (B) Pasteur

Louis Pasteur is credited for developing the rabies vaccine. French scientist Louis Pasteur is well known for his contributions to immunology and microbiology. In 1885, he created the first rabies vaccine, which was successfully used to the treatment of a 9-year-old kid who had been bitten by a rabid dog.

In order to weaken the rabies virus, Pasteur first grew it in rabbits and then dried off the affected nerve tissue. It was possible to immunize people with this attenuated virus and elicit an immunological response without actually infecting them.

The creation of the rabies vaccine by Pasteur was a significant advance in the treatment and prevention of this fatal viral illness, saving countless lives and laying the groundwork for contemporary vaccine research.

While Antonie van Leeuwenhoek and Robert Hooke were early pioneers in the area of microscopy, Edward Jenner discovered the smallpox vaccine in 1796. However, nobody of these researchers created a rabies vaccine.

14. Correct option: (C) Jenner

Edward Jenner was an English physician who is credited with discovering the smallpox vaccine in 1796. He observed that milkmaids who had caught cowpox were immune to smallpox, and he developed a vaccine by inoculating people with cowpox virus. The smallpox vaccine was a major breakthrough in public health, and it is credited with saving millions of lives.

Louis Pasteur developed the rabies vaccine in 1885, Alexander Fleming discovered penicillin in 1928, and Jonas Salk developed the polio vaccine in 1955. However, none of these scientists discovered the smallpox virus.

15. Correct option: (C)

The BCG vaccination (Bacillus Calmette-Guérin) protects against TB. The bacteria Mycobacterium bovis, which is closely linked to the bacterium Mycobacterium tuberculosis, which causes tuberculosis, is the source of the live attenuated vaccination known as BCG. In nations where TB is common, the BCG vaccination is often delivered intradermally to newborns soon after delivery.

It encourages the immune system to identify and react to the TB germs, providing some kind of defense against the disease's more severe forms, especially in youngsters. It is crucial to remember that while the BCG vaccination can lower the chance of developing severe forms of TB, it does not offer full immunity from all tuberculosis infections.
MMR, DPT, and ECG are not TB vaccinations. The MMR vaccination protects against rubella, mumps, and measles. The DPT vaccination guards against tetanus, pertussis, and diphtheria. Electrocardiography, sometimes known as an ECG, is a test that captures the electrical activity of the heart.

16. Correct option: (C)

Acquired Immunodeficiency Syndrome, or AIDS, is brought on by the Human Immunodeficiency Virus (HIV). Specifically targeting CD4+ T cells, which are essential for the body's defense against infections and illnesses, HIV is a retrovirus that affects the immune system.
HIV infection causes a person's immune system to progressively deteriorate over time, increasing their vulnerability to different opportunistic infections and certain malignancies. HIV infection can develop into AIDS, an advanced form of the illness marked by significant immune system dysfunction, if untreated.
Unprotected sexual contact, sharing infected needles, and mother-to-child transmission during delivery or nursing are the main ways that HIV is spread. To control HIV infection and halt the development of AIDS, effective antiretroviral medication (ART) is available, enabling people to live longer, healthier lives.

17. Correct option: (C) Yeast

Option C) Yeast is largely responsible for turning sugars into alcohols. An important part of fermentation activities, including the transformation of carbohydrates into alcohol, is played by yeast, a kind of fungus. A process known as alcoholic fermentation is specifically carried out by yeast, in which the enzymes within the yeast cells break down the sugars into simpler chemicals, chiefly ethanol (alcohol) and carbon dioxide.
This method is frequently used to make alcoholic drinks including beer, wine, and spirits. Yeast is the most well-known and frequently used organism for this purpose due to its effectiveness and predictable fermentation capabilities.
While other bacteria and fungi may also contribute to the conversion of carbohydrates into alcohols in specific situations, this is a function that yeast is best recognized and employed for.

18. Correct option: (D) Mosquitoes

Option D) Mosquitoes are the primary carrier of Dengue and Chikungunya. Both illnesses are spread by the bites of infected members of the Aedes genus of mosquitoes, notably Aedes aegypti and Aedes albopictus. These mosquitoes act as carriers of the viruses that lead to Chikungunya and Dengue. The virus is spread by the saliva of the infected mosquito when it bites a human. The virus subsequently multiplies within the body of the affected person, resulting in symptoms including fever, headache, muscular discomfort, and rash. A mosquito that bites a person who has either virus can pick up the virus and then pass it on to other people it bites.
Numerous diseases are spread by mosquitoes, and in the case of Dengue and Chikungunya, reducing the population of mosquitoes and taking precautions against mosquito bites, such as using insect repellents and donning protective clothing, are crucial for halting the spread of these illnesses.

19. Correct option: (D) 1902

Sir Ronald Ross receives the Nobel Prize in Physiology or Medicine. He was awarded this distinguished honor for his discovery of the mosquito-borne transmission of malaria, particularly the function of Anopheles mosquitoes in this process.

This discovery made a substantial contribution to our comprehension of the malaria transmission cycle and opened the door for further investigation and treatment options for this deadly illness.

20. Correct option: (B) Groundnut

A fungus called tikka disease harms groundnut plants. The fungi Cercospora personata and Cercospora arachidicola are the culprits. Brown patches that form on the groundnut plant's leaves are the disease's primary symptom. These blotches may ultimately combine to produce substantial lesions, which may cause the plant to lose its leaves.

For farmers of groundnuts, tikka disease can be a big issue since it drastically reduces production. The illness may be spread by wind, rain, and insects and is most prevalent in warm, humid settings.

There are several methods for managing the tikka illness, including:

planting resistant groundnut varieties

crops in rotation

burning plant trash with a disease

Fungicide use.

LEVEL-II

1. Correct option: (A) Vibrio cholerae is an example of type S, which causes cholera

Based on their morphologies, bacteria are divided into four main types:

- Spherical (cocci): These microorganisms have an oval or spherical form. E.g., Streptococcus pyogenes.
- Bacilli: These bacteria have a rod form. E.g., Lactobacillus
- Spirilla: These bacteria have a spiral form. Consider the Spirillum minus.
- Vibrios: These bacteria have comma shapes. Eg. Vibrio cholerae.

In the given options correct match is A) Vibrio cholerae is an example of type S, which causes cholera. The bacteria that cause cholera is called Vibrio cholerae. When left untreated, cholera, an acute diarrheal illness, can be fatal and cause severe dehydration. Especially in places with inadequate sanitation and no access to clean water, Vibrio cholerae is primarily spread by contaminated food or water.

The bacteria secrete a toxin that damages the intestines and causes dehydration, vomiting, and the typical cholera symptoms of copious watery diarrhea. To manage cholera patients and stop its spread in impacted populations, quick rehydration and antibiotic therapy are essential

2. Correct option: (A) X - Streptococcus; Y - Aerobic respiration

Yeast from the Saccharomyces genus is frequently employed in the manufacture of wine and alcohol for commercial purposes. It is renowned for its capacity to effectively carry out the fermentation process that turns sugar into alcohol. Brewer's yeast or baker's yeast, sometimes referred to as Saccharomyces cerevisiae, is a well-known species in this genus and is frequently employed in the fermentation of alcoholic beverages.

Through the activity of microorganisms like yeast, fermentation is the process of turning carbohydrates, such as glucose or fructose, into alcohol and carbon dioxide. Alcohol (ethanol) is a result of the yeast's metabolism of the carbohydrates during fermentation. This method is frequently used to make alcoholic drinks as well as other products, such vinegar that contains acetic acid.

3. Correct option: (D) It is a parasitic fungus that causes various diseases in plants, animals, and humans

A filamentous fungus called Rhizopus stolonifer, sometimes referred to as black bread mould, is found on bread surfaces. It gets its nutrition from the bread, which leads to deterioration and destruction. As an opportunistic pathogen, this global mould may

potentially cause zygomycosis infections in people with compromised immune systems. It may result in cutaneous, lung, or sinus infections in certain situations.
Rhizopus stolonifer is also known to produce soft rot in sweet potatoes, which results in dark, squishy regions and an unpleasant odor by decomposing the tissues of the vegetable. In order to avoid contamination and reduce the hazards brought on by Rhizopus stolonifer and other moulds, it is crucial to handle food properly and practice good hygiene.

4. Correct option: (C) A – iv, B – ii, C – i, D – iii

A. Varicella zostera (virus) - iv. Chickenpox
The extremely infectious viral ailment known as chickenpox, which results in painful, blister-like skin rashes, is brought on by the varicella zoster virus.
B. Bordetella pertussis - ii. Whooping cough
Whooping cough is a highly infectious respiratory infection caused by the bacteria Bordetella pertussis, which is characterized by violent coughing fits and a unique "whooping" sound when inhaled.
C. Rhinovirus - i. Common cold
One of the major viruses to blame for the common cold, a moderate viral infection of the nose and throat that frequently includes symptoms like runny nose, sneezing, and congestion, is the rhinovirus.
D. Rubeola virus - iii. Measles
Measles is a highly infectious viral disease that predominantly affects children and is brought on by the Rubeola virus. A red rash, high fever, cough, runny nose, and other flu-like symptoms are its defining features.

5. Correct option: (B) Microorganisms exist in soil, water, air, food, etc.

'Omnipresent' refers to being widely dispersed or present everywhere. In all surroundings, including soil, water, and the air, as well as inside the bodies of plants, animals, and people, microorganisms may be found. By taking part in nutrient cycling, decomposition, and other ecological processes, they perform crucial roles in ecosystems. The term "omnipresent" is thus supported by option B since it accurately depicts the broad existence of microbes in many settings.

6. Correct option: (D) Only 1

Fungi – Eukaryotic, unicellular, or multicellular
Fungi are eukaryotic organisms that can exist as either unicellular (yeast) or multicellular (molds and mushrooms). This statement is correct.
Bacteria – Eukaryotic, multicellular
This statement is incorrect. Bacteria are prokaryotic organisms, meaning they lack a true nucleus and other membrane-bound organelles. They are typically unicellular.
Protozoa - Prokaryotic, parasite
This statement is incorrect. Protozoa are eukaryotic organisms and not prokaryotic. They are unicellular and often free-living, although some can be parasitic.
Virus – Eukaryotic, multicellular
This statement is incorrect. Viruses are acellular particles that are not considered living organisms. They are composed of genetic material (DNA or RNA) enclosed in a protein coat and cannot replicate on their own. They are not eukaryotic or multicellular.

7. Correct option: (D) 1,2 & 3

In reaction to an outside threat, such as a virus or bacterium, the immune system of the body produces proteins called antibodies. They are made to recognize and bind to the foreign material with particularity, which aids the immune system in destroying it.
The first claim is that antibodies are chemical defenses that battle infections. This is true because antibodies are proteins made particularly to attach to and recognize alien objects

like viruses and bacteria. This binding aids in the immune system's ability to eliminate the foreign material.
Statement 2: Proteins make up antibodies. Additionally, since antibodies are proteins, this is true. They are composed of amino acids and are created by B cells in the immune system.
Statement 3: By eliminating the pathogens, antibodies save us from sickness. This is also true since antibodies may assist the immune system via a number of techniques in destroying foreign substances.

8. Correct option: (A) Both A and R are true, and R is the correct explanation for A

The baking sector utilizes yeasts like Saccharomyces cerevisiae because they create carbon dioxide during fermentation. Through thermal expansion, this carbon dioxide makes the bread dough rise. The cause is that when the dough heats up, the stored carbon dioxide gas expands and causes the dough to rise.
As a result, the reason is the proper justification for the statement, and both the assertion and the reason are true.
Assertion: Yeasts such as Saccharomyces cerevisiae are used in baking industry.
Yeasts are single-celled fungi that can ferment carbohydrates to release alcohol and carbon dioxide. Bread dough rises as a result of this fermentation process. Because it is so effective at creating carbon dioxide, the yeast Saccharomyces cerevisiae is frequently employed in baking.
Reason: Carbon dioxide produced during fermentation causes bread dough to rise by thermal expansion.
When yeast ferments sugars, it produces carbon dioxide gas. This gas is trapped in the bread dough, and as the dough heats up in the oven, the gas expands, causing the dough to rise. The thermal expansion of the carbon dioxide gas is what causes the bread to become light and fluffy.

9. Correct option: (C) Both the statement is correct

Statement 1: Vaccines are the weakened disease-causing microorganisms that are injected into our body to develop immunity.
This claim emphasizes the idea that disease-causing bacteria are used in vaccinations in weakened or inactivated forms. In order to trigger an immune response, infections or their components are frequently weakened, destroyed, or injected into the body as part of vaccines. The immune system can learn to identify and create a defense against the genuine disease-causing germs by being exposed to these weaker forms.
Statement 2: Vaccination is the administration of antigenic material to stimulate the immune system of a person to protect against diseases.
In this sentence, the goal and broader definition of vaccination are emphasized. Antigenic material, which can comprise pathogens that have been weakened or rendered inactive as well as certain proteins or genetic components originating from the pathogens, is administered during vaccination. The objective of the immune system stimulation and immunological response is to protect the individual receiving the vaccination from further infections by the disease-causing germs.
In conclusion, Statement 2 offers a broader perspective by emphasizing that vaccination involves the administration of antigenic material to stimulate the immune system and provide protection against diseases, whereas Statement 1 concentrates on the use of weakened disease-causing microorganisms in vaccines.

10. Correct option: (A) Rhizobium bacteria

Rhizobium bacteria are those that support the transformation of atmospheric nitrogen into nitrite or nitrate in conjunction with leguminous plants. These bacteria live in nodules on the roots of leguminous plants in a symbiotic relationship.

Rhizobium bacteria use a process known as nitrogen fixation to transform atmospheric nitrogen into a form that plants may use for growth and development. For the purpose of supplying the soil with nitrogen and enhancing plant health, the mutualistic interaction between Rhizobium bacteria and leguminous plants is crucial.

11. Correct option: (C) Virus

A virus is a type of microbe that can only multiply inside the cells of its host. The unique infectious agent known as a virus needs a host cell to replicate and grow. Outside of a host cell, they are unable to proliferate independently. Once a virus has entered a host cell, it exploits the cellular machinery to manufacture new virus particles, which eventually burst out of the host cell, harming the cell and dispersing the infection. Viruses have a unique mechanism of reproduction that sets them apart from other microbes like diatoms, flagella, and Rhizopus.

12. Correct option: (C) Malaria & Amoebiasis

Malaria and amoebiasis are two of the diseases produced by protozoa.

The protozoan parasite of the genus Plasmodium, which causes malaria, infects female Anopheles mosquitoes and spreads to people by their bites.

Entamoeba histolytica is a protozoan parasite that causes amoebiasis, often known as amoebic dysentery. It typically spreads when contaminated food or water is consumed.

The other diseases mentioned in the options have different causative agents:

- Smallpox is caused by the variola virus.
- Tuberculosis is caused by the bacterium Mycobacterium tuberculosis.
- Typhoid is caused by the bacterium Salmonella typhi.
- Chikungunya is caused by the chikungunya virus.

13. Correct option: (D)

Since most microorganisms are so tiny, they cannot be seen with the human eye. They generally range in size from a few micrometers to submicron or nanometer scale, making them tiny. Only a microscope will allow you to see them.

Both "microorganisms are helpful" and "microorganisms are dangerous" are true. Depending on the species, microorganisms can either be advantageous or harmful. For instance, while certain microbes are capable of causing disease, others can aid in the digestion of food and the production of vital nutrients.

Therefore, "microorganisms can be seen by the naked eye" is the sole claim that is false.

14. Correct option: (D) Statement 3 is correct only

Statement 1: Fishes are preserved by adding sugar.

This is incorrect. Fish can be preserved via smoking, drying, or salting.

Sugar draws moisture, which makes it a bad preservative for fish since it might inducerot ting.

Statement 2: Jams and jellies are preserved by adding salt.

Sugar is used to preserve jams and jellies.

Jams and jellies shouldn't be preserved with salt since it could impart a salty flavor.

Statement 3: Vegetable slices are preserved by adding salt and drying in sunlight.

This is correct. By salting vegetable slices and letting them dry in the sun, you may preserve them. Vegetables are dried out by the salt, which helps keep them from rotting. Any microbes that might be on the veggies are also assisted by the sunshine in their destruction.

15. Correct option: (C) W – Virus, X – Fungus, Y – Protozoan

The microorganisms are classified based on their characteristic structures

W – Virus - The cells of the host organism are the sole area where viruses may reproduce. Viruses are contagious, non-living creatures. They are composed of genetic material (DNA or RNA) covered with proteins called a capsid, which is composed of tiny units called capsomeres. These capsomeres are structured geometrically in helical or polyhedral shapes.

X – Fungus - Eukaryotic creatures called fungi take nutrients from their environment by absorbing them. They may reproduce through spores and are distinguished by the filamentous structures they possess, known as hyphae.
Y – Protozoan - Eukaryotic single-celled creatures called protozoa exist. They fall under the protist kingdom classification and are found in a variety of habitats. The different forms and means of protozoa's motility, such as flagella, cilia, or pseudopodia, are well recognized.

16. Correct option: (A) P – Air, Q – Tuberculosis, R – Bacteria, S - Malaria

P is spread by the air, and a virus is to blame for the disease. Viral illnesses like measles and mumps can be transmitted by respiratory droplets.
Q is the tuberculosis disease which is spread by the air, and the bacterium Mycobacterium tuberculosis that causes airborne illness TB.
R is bacteria Salmonella typhi that may spread by tainted food and water that causes typhoid fever.
S is the malaria disease spread by mosquitoes, which act as vectors, and a protozoan causes the disease. The protozoan parasite of the genus Plasmodium is the root cause behind malaria, an illness spread by mosquitoes.

17. Correct option: (B) It is a type of virus which attacks bacteria

A bacteriophage, sometimes referred to as a phage, is a kind of virus that infects and spreads throughout bacteria. It targets bacteria selectively and reproduces by using them as host cells. Bacteriophages are multi-layered organisms that contain genetic material (DNA or RNA) that is covered in a protein coat.
In order to make additional phages, they adhere to the surface of bacterial cells, inject their genetic material there, and take over the host's cellular machinery. Bacteriophages eventually cause infected bacterial cells to lyse (break open), which releases fresh phages that can infect further bacterial cells.

18. Correct option: (D) 1, 2 & 3

To avoid food poisoning, food must be kept properly: The growth of bacteria and other pathogens that might result in food poisoning is slowed down by proper food storage practices, such as refrigeration at the right temperatures.
Longer periods of time without refrigeration can cause food poisoning: Bacteria may quickly develop when perishable foods are kept at room temperature for a lengthy period of time, increasing the risk of foodborne diseases.
Food handling errors can also result in food poisoning. For example, improper hand hygiene, mixing raw and cooked meals, and using contaminated utensils or surfaces can all lead to the introduction of bacteria into the food.

19. Correct option: (C) b only

Autotrophs are creatures that have the ability to synthesize their own food via chemosynthesis or photosynthesis. Algae are photosynthesis-capable creatures that can absorb solar energy and use photosynthesis to transform it into chemical energy. They make use of this energy to create organic compounds like carbohydrates, which are their main food supply.
An oxygen-free kind of respiration is called anaerobic respiration. Algae do not require anaerobic respiration since they do not utilize it.
Heterotrophic feeding occurs when an organism consumes another live thing as well as dead ones. Since algae are autotrophs, they don't require a source of food that is either living or dead.

20. Correct option: (B) (i) - (e); (ii) - (c); (iii) - (a); (iv) - (b); (v) - (d)

i. Louis Pasteur made significant contributions to the field of microbiology and fermentation. In 1857, he conducted experiments that demonstrated the role of microorganisms in the process of fermentation.

ii. Alexander Fleming discovered the antibiotic substance known as penicillin in 1929. His discovery revolutionized the field of medicine by introducing the first widely used antibiotic.
iii. Edward Jenner is credited with developing the first successful smallpox vaccine in 1798. His work laid the foundation for the field of immunization.
iv. Robert Koch was a renowned microbiologist who identified the bacterium Bacillus anthracis as the causative agent of anthrax in 1876.
V Anton van Leeuwenhoek was a Dutch scientist who is considered the father of microbiology. He was the first to observe and describe microorganisms using a simple microscope, which he designed and built himself

21. Correct option: (A) Only 2

Statement 2 states that Chikungunya is an infectious disease caused by bacteria. However, this statement is incorrect. The Chikungunya virus, which is a member of the Togaviridae family, is what really causes the disease.
Chikungunya is spread to people through the bite of infected mosquitoes, especially Aedes mosquitoes, which are also known to spread other viral illnesses including dengue and Zika. When the Chikungunya virus infects someone, they may get symptoms including a high temperature, joint discomfort, rash, headache, and muscle soreness.

22. Correct option: (D) A, B & D

A) A single-celled fungus called yeast is utilized in fermentation. Alcohol and carbon dioxide are produced from sugar during the fermentation process. Bread, beer, wine, and other foods that go through the fermentation process all utilize yeast. Therefore, it is a correct match.
Bacteria that can fix nitrogen include blue-green algae. The process of nitrogen fixation involves transforming atmospheric nitrogen into a form that plants can utilize. Due to their ability to enhance the quantity of nitrogen in the soil, blue-green algae are crucial for soil fertility. Therefore, it is a correct match.
A bacteria called Rhizobium collaborates symbiotically with legumes. In this interaction, a bacterium that fixes nitrogen resides in the root nodules of legume plants. The legume plant then has access to this nitrogen, aiding in the improvement of the plant's growth. Therefore, it is an incorrect match
The fungus penicillium is responsible for making penicillin. Several different bacterial infections can be treated with penicillin. Therefore, it is a correct match.

23. Correct option: (B) C, D, A and B

C)Dig a pit - This is the initial stage of the manure-making process. The pit needs to be large enough to hold all of the organic waste that will be added, and it needs to be excavated in a well-drained region.
D)Dump the organic waste into pit - Food scraps, animal dung, and other organic elements are examples of this. To help keep the waste wet and stop it from luring pests, it should be covered with a layer of dirt.
A) Action of microorganisms - Microorganisms will eventually decompose the organic waste. The waste will be transformed into manure by these microbes, which is a great source of nutrients for plants.
B) Formation of manure - The manure will be ready to use after a few months. It can be applied to gardens and lawns to improve soil fertility.

24. Correct option: (A) a, c and d

a) Antibiotics are manufactured from fungi and bacteria: In 1928, Alexander Fleming isolated the first antibiotic, penicillin, from a fungus known as Penicillium notatum. Since then, several additional antibiotics derived from fungus and bacteria have been found.
b) Antibiotics are synthesized in our body: Our bodies cannot produce antibiotics. They are made by other living things like fungus and bacteria.

c) Antibiotics are not effective against viral diseases: Bacteria are either killed or have their growth slowed by antibiotics. As viruses are a distinct kind of creature, they are ineffective against them.
d) Antibiotics are mixed with the feed of livestock: In order to stop the transmission of illnesses among animals, this is done. Antibiotic resistance in bacteria has, however, been related to the use of antibiotics in cattle feed.

25. Correct option: (C) D, B, A and C

1. Food spoilage involves breakdown of food by toxic enzymes. This is the first step in the process of food spoilage. The enzymes are produced by microorganisms, such as bacteria and fungi, that are present in the food.
2. Evolution of carbon dioxide. As the food spoils, the microorganisms will produce carbon dioxide. This carbon dioxide will cause the food to become bloated and mushy.
3. Food becomes stale, smelly, mushy. The food will also become stale, smelly, and mushy as it spoils. This is due to the breakdown of the food by the microorganisms.
4. Stale food causes illness. If stale food is eaten, it can cause illness. This is because the microorganisms that have spoiled the food can also produce toxins that can make people sick.

26. Correct option: (A) Both A and R are true, and R is the correct explanation for A.

In fact, the symbiotic relationship between algae and fungus produces lichens.
While the fungus creates a protective framework and may absorb water and nutrients, th e algae contribute photosynthetic powers and food production for the partnership.
Lichens may exist and flourish in a variety of habitats thanks to their mutualistic interact ion.
The explanation (R), which emphasises the distinct functions that both algae and fungus play in this interaction, also adequately explains why lichens include a symbiotic associat ion.
The proper response is therefore A) Both A and R are true, and R is the appropriate justifi cation for A.

27. Correct option: (A) (i) → (e); (ii) → (a); (iii) → (c); (iv) → (b); (v) → (d)

(i) Canning - (e) Sealed food in metal: Food is preserved via canning, which involves sealing it in airtight vessels, usually made of metal. The longer the food is preserved, the more protection against air and bacteria it receives from this procedure.
(ii) Chemical preservative - (a) Vinegar: Chemical preservatives are compounds that are applied to food to stop microbial development and lengthen shelf life. Due to its antibacterial qualities, vinegar, which is acidic, functions as a natural chemical preservative.
(iii) Pasteurization - (c) Heat and cold treatment: In order to pasteurize food or drink, it must first be heated to a certain temperature and then quickly cooled. It also helps to prolong the shelf life of the items by killing or inactivating dangerous microorganisms.
(iv) Pickling - (b) Salt and oil: Food is submerged in a solution containing salt, vinegar, or brine as part of the pickling preservation technique. This method aids in establishing conditions that prevent bacterial development and increase the food's shelf life. Pickling can also use oil for flavor and further preservation.
(v) Freeze drying - (d) Froze ice to vapor: Freeze drying, sometimes referred to as lyophilization, is a method of food preservation that entails freezing the food and then sublimating the ice to remove it. This method considerably increases the food's shelf life while maintaining the food's natural flavor, texture, and nutritional content.

28. Correct option: (C) III and IV

Statement I: is correct: An individual's immunity is their capacity to fend off illnesses. This is accomplished through the immune system's coordinated reaction to any foreign particles.

Statement II: is correct: The formation of antibodies, the activation of white blood cells, and the release of chemicals that might destroy or weaken foreign particles are just a few of the processes that may be involved in this response.
Statement III: is incorrect: Cyanobacteria are the bacteria that encourage curd production.
Actually, Lactobacillus, not Cyanobacteria, is the bacterium that aids in the production of curd.
Photosynthetic microorganisms called cyanobacteria are typically found in watery habitats.
Statement IV: is incorrect: Spirogyra sp. is a type of algae, not a fungus. Algae are photosynthetic organisms that are found in both freshwater and saltwater environments. Fungi are heterotrophic organisms that decompose dead organic matter.

29. Correct option: (A) X – Bacteria, Y – Fungi, Z – Virus

Citrus canker: A bacterial condition known as citrus canker is brought on by Xanthomonas axonopodis pv. citri. The bacteria can spread through the water, the air, or plant matter that has been contaminated.
Rust of wheat: Puccinia graminis is the fungus that causes wheat rust. The fungus can spread through the air or on plant material that has been contaminated.
Yellow vein mosaic of bhindi: The cucumber mosaic virus is the cause of the viral illness known as yellow vein mosaic of bhindi. Infected seeds, mechanical devices, or insects can all spread the virus.

30. Correct option: (C) Only B

Due to their capacity to prevent bacterial development, oil and vinegar are frequently employed as preservatives. Food preservation is made possible by the fact that both oil and vinegar foster an environment that is unfavourable for bacterial development. While oil serves as a barrier to keep germs from reaching the food and reducing oxygen exposure, the acidity of vinegar can produce an unfavorable habitat for some bacteria.

3. Conservation of Plants and Animals

Solutions

LEVEL-I

1. Correct option: (C) Endangered Species

The species very near to extinction, if conservational measures are not prompt are called endangered species.

2. Correct option: (A) International union for conservation of nature and natural resource

The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of: 1. Endemic plant and animal species present in the biodiversity hotspots.

3. Correct option: (C) Both A and B

Wildlife is conserved by both in situ and ex situ methods.

In situ or on-site conservation involves the conservation of genetic resources of natural populations of plant and animal species, such as from forests, National Parks, and Biosphere Reserves.

Ex situ conservation involves the preservation of components of biological diversity outside their natural habitats, such as in Zoological Parks.

4. Correct option: (B) 50000

India has more than 50000 genetically different strains of rice.

5. Correct option: (A) Biosphere reserve

Biosphere reserves are large areas of protected land for conservation of wildlife, plant and animal resources and traditional life of the tribals living in the area. The biosphere reserves help to maintain the biodiversity and culture of that particular area.

6. Correct option: (D) All

The fundamental purpose of plant gene banks is to preserve genetic varieties in the form of seeds or cuttings for plants that reproduce vegetatively, and subsequently, make them accessible for use in plant breeding and research in the future.

7. Correct option: (C) Habitat destruction

Habitat loss is the primary cause of higher extinction rates. Other causes include habitat changes, over-exploitation of wildlife for commercial purposes, the introduction of harmful nonnative species, pollution, and the spread of diseases.

8. Correct option: (D) Threatened species

The IUCN Red List provides information about the geographic range, population size, habitat and ecology, use and trade, threats and conservation actions for species.

9. Correct option: (B) Fossil Fuel

Coal, natural gas and crude oil are called as fossil fuels. They are regarded as non-renewable source because it takes millions of years to form and once these resources are used, they cannot be regained.

10. Correct option: (C) Increasing Humidity and rainfall

Forests helps in producing rainfall. More are the trees; the more water gets absorbed into clouds and the more it helps in rain falls. The roots of a tree make gaps in the soil so that when it rains the water can sink in before being absorbed by the soil. These spaces also allow air and water to reach the roots of the plants. If the forests disappear, there will be less rain resulting in dryer conditions, that eventually lead to drought. Once this happens, the damage is extensive making it difficult to plant more trees or plants later because the soil is so hard and dry.

11. Correct option: (A) Jatropha

Biofuels are of two types in which bioethanol is one of the types. Jatropha plant is used as an alternative to produce bioethanol. A high amount of energy is produced by Jatropha as its seeds are rich in fatty acid methyl ester. Hence the biofuel bioethanol can be yielded by processing the oil obtained from Jatropha seeds. Its oil is mixed with pongamia oil to produce bioethanol.

12. Correct option: (C) East Africa

Lake Victoria, also called Victoria Nyanza, largest lake in Africa and chief reservoir of the Nile, lying mainly in Tanzania and Uganda but bordering on Kenya.

13. Correct option: (C) Lungs of Planet

Amazonian rain forest in south America has the greatest biodiversity on earth; it harbors about40000 species of plants, 1,25,000 species of insects, 3000 species of fishes, 427 of amphibians, 378 of reptiles, 1300 of birds & 427 of mammals.

14. Correct option: (A) Endemic Species

Endemic species are plants and animals that exist only in one geographic region. Species can be endemic to large or small areas of the earth: some are endemic to a particular continent, some to part of a continent, and others to a single island.

15. Correct option: (D) Botanical Garden

These include aquariums, botanical gardens, gene banks etc. In Situ conservation refers to the conservation of the plants and animals in their own habitat. These include biosphere reserves, wildlife sanctuary and national parks.

16. Correct option: (A) Endemic population

The population limited to a particular geographical area is called an endemic population.

17. Correct option: (C) Root of rauwolfia serpentine

Reserpine is an indole alkaloid extracted from Rauwolfia serpentine roots, an Indian climbing shrub.

18. Correct option: (C) Gene bank

Endangered plant species are conserved through Gene bank. By this method, different genetic diversity is preserved and thus creates a genetic pool of different genetic material that acts as a resource house.

19. Correct option: (C) Cryopreservation

Germplasm conservation at liquid nitrogen temperature is called cryopreservation. It is an example of Ex-situ conservation. Ex-situ conservation is done by preserving the components of biological diversity outside their natural habitats.

20. Correct option: (A) Pollution

UNESCO's Biosphere Reserves face threats such as habitat loss, climate change impacts, invasive species, pollution, unsustainable resource use, and development pressures.

21. Correct option: (B) In situ conservation

The conservation of species in their natural habitat or natural ecosystem is known as in-situ conservation. In the process, the natural surrounding or ecosystem is protected and maintained so that all the constituent species (known or unknown) are conserved and benefited.

22. Correct option: (D) 29 December

From its creation by the Second Committee of the UN General Assembly in 1993 until 2000, it was held on December 29 to celebrate the day the Convention on Biological Diversity went into effect. On December 20, 2000, the date was shifted to commemorate the adoption of the Convention on May 22, 1992, at the Rio de Janeiro Earth Summit, and partly to avoid the many other holidays that occur in late December.

23. Correct option: (D) −196°C

Cryopreservation is the conservation of living cells, tissues organs, and microorganisms at ultralow temperature (usually that of liquid nitrogen, −196°C).

24. Correct option: (B) Maximum biodiversity

Biodiversity hotspots are the regions of megadiversity. These areas are rich in biodiversity. They reserve the endangered species.

25. Correct option: (B) Deforestation

Deforestation is the depletion of forest resources. Its main cause is the explosion of human and livestock population with the increased demand requirements of the basic needs. The major effect of deforestation is the loss of precious wildlife, rare species of flora and fauna. Directly or indirectly, deforestation caused intensified soil erosion, accentuated floods, drought and the worst pollution.

26. Correct option: (C) Contains diverse genetic resources

All life forms should be conserved because, they maintain diverse genetic resources.

27. Correct option: (B) 2.4%

India occupies 2.4% of the world's land area with its 3.28 million square kilometre area.

28. Correct option: (D) Coexistence

There are four major causes collectively called as the evil quarter- i) Overexploitation, ii) Loss of habitat, iii) Extinction of species, iv) Introduction of exotic species. Thus Co – Existence is not included in the Evil Quartet for accelerated rates of species extinction.

29. Correct option: (B) Alkaloids

Drugs obtained from plants are mostly alkaloids in nature. Examples: - Quinine obtained from the bark of Cinchona stem. Taxol obtained from the bark of Pacific yew. Reserpine obtained from Rauvolfia serpentina. Morphine obtained from the poppy plant.

30. Correct option: (D) Seed Bank

Seed banks are the conservation site where seeds are stored for conserving species as seeds can stay in a dormant stage for a longer span until it gets favorable conditions. As plants are conserved by this method outside their habitat, it is a type of Ex-situ conservation.

LEVEL II

1. Correct option: (B) Increased resistance to environmental perturbance

Biodiversity loss is defined as the extinction of species worldwide, as well as the local reduction or loss of species in a certain habitat, resulting in a loss of biological diversity. Loss of biodiversity may lead to decline in plant production, increased variability in plant productivity, Increased variability in water use and in pest and disease cycle. There would also be a decrease in resistance to environmental disturbances.

2. Correct option: (C) 1952

The National Forest Policy in India was enforced in 1952 by the government of India.

3. Correct option: (D) Red Data Book

The 'Red Data Books' published by the International Union for Conservation of Nature and Natural Resources (IUCN) contain lists of: 1. Endemic plant and animal species present in the biodiversity hotspots.

4. Correct option: (A) Soil erosion

Global warming is the rise in the temperature of the earth's atmosphere. This results in the melting of polar glaciers, which increases the sea level, causing floods in the coastal areas. Global warming can also cause an increase in the temperature of the ocean, which causes the death of marine flora and fauna. Global warming has no effect on soil erosion.

5. Correct option: (A) 1992

The United Nations Conference on Environment and Development (UNCED), also known as the 'Earth Summit', was held in Rio de Janeiro, Brazil, from 3-14 June 1992.

6. Correct option: (D) All of these

A wildlife sanctuary is an area where animal habitats and their surroundings are protected from any sort of disturbance. The capturing, killing, and poaching of animals is strictly prohibited in these regions. They aim at providing a comfortable living to the animals.

7. Correct option: (B) On 1 April 1973

The Government of India launched the Project Tiger with the aim of saving the steadily declining population of tigers in India on 1 April 1973. Project Tiger has been the largest species conservation initiative of its kind in the world.

8. Correct option: (A) Bori and Pachmarhi

The Panchmarhi Biosphere Reserve consists of one National Park named Satpura and two wildlife sanctuaries named Bori and Panchmarhi.

9. Correct option: (C) Wildlife sanctuary and National Park

Wildlife sanctuary and national parks are the places meant for conservation of biodiversity in their natural habitat.

10. Correct option: (A) Pachmarhi Biosphere Reserve

The biodiversity found in the Panchmarhi Biosphere Reserve is similar to that found in the upper Himalayan peaks and lower Western Ghats.

11. Correct option: (D) All of the above

Forests provide us with oxygen, shelter, jobs, water, nourishment and fuel.

12. Correct option: (C) Is occurring at a faster rate

Unlike previous extinction events caused by natural phenomena, the sixth mass extinction is driven by human activity, primarily (though not limited to) the unsustainable use of land, water and energy use, and climate change, and is occurring at relatively higher rates than the previous ones.

13. Correct option: (A) Habitat loss and fragmentation

When a habitat is destroyed, the carrying capacity for indigenous plants, animals, and other organisms is reduced so that populations decline, sometimes up to the level of extinction. Habitat loss is perhaps the greatest threat to organisms and biodiversity.

14. Correct option: (D) Biodiversity

Biodiversity refers to the variety of organisms existing on the earth, their interrelationships, and their relationship with the environment. Biodiversity or Biological diversity is a term that describes the variety of living beings on earth. It helps in maintaining the balance of the ecosystem.

15. Correct option: (A) Endemic species

Endemic species are plants and animals that exist only in one geographic region. Species can be endemic to large or small areas of the earth: some are endemic to a particular continent, some to part of a continent, and others to a single island.

16. Correct option: (B) Preservation of species

Creation of wildlife reserves and enforcement of laws are conservation measures that promote increased preservation of species.

17. Correct option: (C) Hunting and Poaching

The illegal hunting and poaching of animals can lead to a decrease in their numbers, which can have negative effects on the overall ecosystem. Poaching of endangered species such as elephants, tigers, and rhinos can lead to their extinction, causing irreversible damage to the environment.

18. Correct option: (D) All of the above

The species which no longer exist anywhere on the earth are called extinct species. The species which are facing the risk of extinction are called endangered species. Red data book keeps a record of all the endangered plants and animals or the species which are at continuous threat of becoming extinct. There are various colour codes for various types of species depending on the level of risk they are in when it comes to becoming extinct.

19. Correct option: (A) Habitat Destruction

Habitat loss is the primary cause of higher extinction rates. Other causes include habitat changes, over-exploitation of wildlife for commercial purposes, the introduction of harmful nonnative species, pollution, and the spread of diseases.

20. Correct option: (C) Very cold

Migratory birds fly to far away areas every year during a particular time because of climatic changes. They fly for laying eggs as the weather in their natural habitat becomes very cold and inhospitable.

21. Correct option: (D) Interbreeding

A species is a group of populations which are capable of interbreeding to produce a fertile offspring.

22. Correct option: (D) Habitat

A place where an organism or a community of organisms live, including the living and non-living conditions of their surrounding environment, is called a habitat.

23. Correct option: (A) Animals move freely inside a national park

In any national park, the animals and plants are conserved in their natural habitat. The wild animals roam freely. Hence, tourists must watch them through a well-protected vehicle for their own safety.

24. Correct option: (B) Decrease in floods

Increase in groundwater level. An increase in groundwater level is not a consequence of deforestation. It is a very broad term, which consists of cutting of trees including repeated lopping, felling, and removal of forest litter, browsing, grazing, and trampling of seedlings.

25. Correct option: (B) Biosphere

Part of the earth which supports life where living beings exists is called biosphere. The biosphere is the layer of the planet Earth, where life exists.

26. Correct option: None are correct (explanation below)

According to IUCN Red list, Labrador duck is an extinct species. Cheetah and leopard are vulnerable species whereas Jaguar is near threatened species. Hog dear, red panda, Asian elephant and blue whale are endangered species.

27. Correct option: (A) Dodo

Dodo is an extinct flightless bird that was endemic to the island of Mauritius, east of Madagascar in the Indian Ocean. It became extinct due to over-exploitation by human beings.

28. Correct option: (B) a-ii; b-iii; c-i

Dodo is an extinct flightless bird that was endemic to the island of Mauritius. Giant squirrels are Endemic to Western, Eastern Ghats and Satpura range of India. Tigers have lost an estimated 95% of their historical range. Their habitat has been destroyed, degraded, and fragmented by human activities. Hence, they are categorised as endangered animals by IUCN.

29. Correct option: (D) 1-b; 2-d; 3-a; 4-c

Giant squirrels are Endemic to Western, Eastern Ghats and Satpura range of India. Cutting of trees in the forests is known as deforestation. A food chain is a linear sequence of organisms through which energy and nutrients pass as one organism eats another. Pachmarhi comprises three protection sites: the Bori Sanctuary, Satpura National Park and Pachmarhi Sanctuary – otherwise known as the Satpura Tiger Reserve.

30. Correct option: (D) (i), (ii), (iii), (iv)

Deforestation is removal, decrease or deterioration of forest cover of an area. It causes shrinkage of fuelwood, reduced timber supply, change in climate, flash floods, soil erosion, siltation, cyclones, drought, desertification, reduced rainfall and destruction of habitat.

4. Cell - The Fundamental Unit of Life

Solutions

LEVEL-I

1. Correct option: (A) Oak tree
Cork has dead cells which resembles the cells found in Matchstick. Onion peel cells and Cheek cells are living cells.
2. Correct option: (B) Robert Brown
Robert Brown discovered nucleus in 1831.
3. Correct option: (D) Nucleus
Robert Brown observed nucleus in a plant cell while studying the process of fertilization and claimed that it is an integral part of any cell, in the year 1831.
4. Correct option: (A) Orchid
Robert Brown coined the term 'nucleus' in 1831. He first observed the nucleus in the epidermis of the orchids as an opaque spot.
5. Correct option: (A) Cytoplasm
Cytoplasm is the jelly-like substance present between the cell membrane and the nucleus. It is mainly composed of water, salts, and proteins. All the membrane bound cell organelles like endoplasmic reticulum, golgi complex, etc are present in the cytoplasm.
6. Correct option: (D) All of the above
Unicellular organisms include all prokaryotes including Bacteria, and Eukaryotes including Chlamydomonas sp, and Amoeba sp.
7. Correct option: (C) Amoeba sp.
Pseudopodia or false feet are cytoplasmic extensions present in unicellular protists, and functions in locomotion and capturing of prey.
8. Correct option: (A) Protection
The dermal tissue covers the outer surface of the stem and usually functions to waterproof, protect, and control gas exchange.
9. Correct option: (D) All of these
Any process that is commonly exhibited by living organisms on a daily basis for carrying out daily metabolic processes can be termed as a life process.
10. Correct option: (C) Janssen
A Dutch father-son team named Hans and Zacharias Janssen invented the first so-called compound microscope in the late 16th century when they discovered that, if they put a lens at the top and bottom of a tube and looked through it, objects on the other end became magnified.
11. Correct option: (D) Cell
The cell is the basic structural and functional unit of life. Cells are independent, single-celled organisms that take in nutrients, excrete wastes, detect and respond to their environment, move, breathe, grow, and reproduce.
12. Correct option: (C) Microscope
Microscope, instrument that produces enlarged images of small objects, allowing the observer an exceedingly close view of minute structures at a scale convenient for examination and analysis.

13. Correct option: (D) Sclerenchyma cells

Mature sclerenchyma cells are usually dead cells that have heavily thickened secondary walls containing lignin. The cells are rigid and non-stretchable and are usually found in nongrowing regions of plant bodies, such as the bark or mature stems.

14. Correct option: (A) The number of cells present

The size of an organism primarily depends upon the number of cells in its body. When the number of cells increases then the size of the organism also increases.

15. Correct option: (A) Cell wall

The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress.

16. Correct option: (D) Both A and B

The functions of pseudopodia include locomotion and ingestion: Pseudopodia are critical in sensing targets which can then be engulfed; the engulfing pseudopodia are called phagocytosis pseudopodia.

17. Correct option: (A) 1-A; 2-D; 3-B; 4-C

Robert Hooke in the year 1665 discovered cell in cork tissue. Robert brown discovered nucleus in orchids in the year 1831. Antonie van Leuwenhoek made the first microscope and visualized live cells. A plant cell is a living cell due to a proper functioning metabolism.

18. Correct option: (D) 2 only

Robert Brown was the one who discovered nucleus while studying orchid leaves and coined the term 'nucleus' in 1831.

19. Correct option: (D) Both A and B

The shape of the human red blood cell is known to be a biconcave disk.

20. Correct option: (D) 90-100 cm

The nerve cell is the structural and functional unit of the nervous system. They are generally long and branched cells and can be 90 - 100 cm long.

LEVEL II

1. Correct option: (B) Cytoplasm

The cytoplasm is a thick, clear, jelly-like substance present inside the cell membrane. Most of the chemical reactions within a cell take place in this cytoplasm. The cell organelles such as endoplasmic reticulum, vacuoles, mitochondria, ribosomes, are suspended in this cytoplasm.

2. Correct option: (D) Cell wall, Chloroplast, Mitochondria, Cell membrane, Nucleus

The following are the components of animal cells: Nucleus, Ribosomes, Endoplasmic reticulum, Lysosomes, Golgi bodies, Cell membrane, Cell wall, nuclear membrane, Mitochondria.

3. Correct option: (C) c and d

As red blood cells contain no nucleus, protein biosynthesis is currently assumed to be absent in these cells. Since its an animal cell it doesn't have a plastid.

4. Correct option: (C) b and d

A nerve cell or neuron is so long about 100 cm so that it can carry messages over long distances from brain to other parts of the body. It has numerous projections to make contact with other neurons.

5. Correct option: (B) Euglena, Chlorella, Escherichia

Euglena is a unicellular protist, Chlorella is an unicellular algae growing in fresh water. Escherichia coli is a gram-negative bacterium hence an unicellular organism.

6. Correct option: (C) A-ii, B-iv, C-I, D-iii

A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.

7. Correct option: (A) Mitochondria - synthesize the respiratory enzymes

Mitochondrion is the site for aerobic respiration in cells. It contains respiratory enzymes necessary for ATP synthesis. A nucleus is the membrane-enclosed organelle within a cell that contains the chromosomes. Cytoplasm is the gelatinous liquid that fills the inside of a cell. It is composed of water, salts, and various organic molecules. A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.

8. Correct option: (B) X – Plastid, Y – Chloroplast, Z – Chromoplast

The chloroplast is the photosynthetic plastid, and it is named for its green colour. Based on pigmentation, the non-photosynthetic plastids can be broadly divided into leucoplasts, the 'white' or colourless plastids, and chromoplasts, the coloured plastids notable for their accumulation of carotenoids.

9. Correct option: (A) a and b only

Nucleus is the largest cell organelle and controls the activities of the cell. The living material of a cell that is made up of both nucleus and cytoplasm is called protoplasm. Chloroplasts are only one, albeit the most prominent, member of a larger family of plant organelles called plastids. All plastids contain the same genome as chloroplasts, but they differ in both structure and function.

10. Correct option: (D) iii and iv only

Plant cells have a cell wall, chloroplasts and other specialized plastids, and a large central vacuole, whereas animal cells do not.

11. Correct option: (A) These are composed of tiny spherical structures surrounded by a double membrane

Lysosomes are solitary membrane-bound organelles that contain digestive enzymes that break down food scraps, bacteria, macromolecules, worn-out organelles, and viruses that have been eaten. Lysosomes do not have a double membrane. They are also called suicide bags of cell as they can digest their own cell.

12. Correct option: (A) i, iii and iv only

Anything inside the plasma membrane is called a protoplasm. This term was firstly introduced by J. E. Purkinje. Rudolf Virchow gave the statement *'Omnis cellula-e-cellula'*. It means "All cells arise from pre-existing cells. The Cell Theory was first given by M.J Schleiden and Theodor Schwann in 1839.

13. Correct option: (B) The egg cell has half as many chromosomes as the skin cell

Egg and sperm cells have just 23 chromosomes each. That's half as many chromosomes as regular cells. Through the process of fertilization, egg and sperm join to make a cell with 46 chromosomes (23 pairs), called a zygote.

14. Correct option: (D) A pancreatic cell engaged in the production of digestive enzymes

Endoplasmic reticulum helps in the transport of substances within the cell. Rough endoplasmic reticulum (RER) bears ribosomes on the cytoplasmic surface and synthesises secretory and membranous proteins. Then the modified proteins move into the transitional area where they are further processed and packed into secretory vesicles. Thus, RER is rich in cells which are actively engaged in protein synthesis and secretory activity, e.g., pancreatic cells engaged in the production of digestive enzymes, etc.

15. Correct option: (B) X could be Amoeba whereas Y could be Mycoplasma

An amoeba is an eukaryotic unicellular cell, that have a well-organized nucleus which is separated from the cytoplasm by the nuclear membrane. Mycoplasmas are the smallest bacterial cells that do not have a true nucleus and their nuclear material is in direct contact with cytoplasm, also they do not have a cell wall.

16. Correct option: (D) Leucoplasts help in the storage of Food

Plastid is the organelle that found only in plant cells. The three types of plastids are green plastid, nongreen chromoplast, and leucoplasts. The leucoplasts are the non-colored plastid

which is used for the storage of the food. The oil, starch, and fats are generally stored in these plastids.

17. Correct option: (C) Only D

Chloroplast are involved in synthesis of food for plant through photosynthesis. Nucleus is a central organelle responsible for controlling all metabolic reactions and contains genetic material of the cell. The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress. Mitochondria is involved in production of energy in the form of ATP.

18. Correct option: (D) Only II

A vacuole is a membrane-bound cell organelle. In animal cells, vacuoles are generally small and help sequester waste products. In plant cells, vacuoles help maintain water balance. Sometimes a single vacuole can take up most of the interior space of the plant cell.

19. Correct option: (C) A is true but R is false

The endoplasmic reticulum which lacks ribosomes is called smooth endoplasmic reticulum and is involved in lipid biosynthesis.

20. Correct option: (A) Golgi complex

Golgi complex receives the substances synthesized and release by the ER, condenses, modifies, packs and releases them in the form of secretory vesicles.

21. Correct option: (C) A – Incoming transport vesicle, B – Lumen, C - Newly form vesicle, D – Cisternae, E –Secretory vesicle

Golgi apparatus also called Golgi complex and found universally in both plant and animal cell. It is comprised of a series of membrane-covered sacs called cisternae that look something like a stack. The cis face of a Golgi stack is also known as forming a face which has a convex face. The trans face is known as the maturing face which has a concave face. From the forming face, several vesicles are pinched off forming.

22. Correct option: (C) Both the statements are correct

Antonie van Leeuwenhoek was the first to see living bodies under a Microscope in the year 1674. Robert Brown discovered the nucleus in the Orchid leaf in the year 1831.

23. Correct option: (A) D, C, B, A

Take a scaly leaf of onion and pull a transparent peel. Place the peel onto a drop of water on a glass slide. Add a drop of methylene blue and place a coverslip. Observe the slide under a microscope.

24. Correct option: (D) A – v, B – iii, C – i, D – ii, E – iv

Onion cell is plant cell that have a cell wall surrounding its membrane. Chromosomes are strands of DNA interwound together and forms a distinct mass during cell division. Methylene blue is a bluish coloured stain. Microscope is an instrument that magnifies and aids in visualization of tiny creatures. Ribosomes are protein manufacturing organelles that maybe freely suspended in cytoplasm or maybe attached on the membrane of rough endoplasmic membrane.

25. Correct option: (D) i, ii, and iii

In 1972, Singer and Nicolson proposed the fluid mosaic model of the plasma membrane. The fluid mosaic model describes the plasma membrane structure as a mosaic of components -including phospholipids, cholesterol, proteins, and carbohydrates. The proportions of proteins, lipids, and carbohydrates in the plasma membrane vary with the cell type.

26. Correct option: (A) Both A and R are true, and R is the correct explanation of A

Chloroplasts are the green plastids. They contain a green coloured pigment called chlorophyll. Chlorophyll is the key component in the process of photosynthesis which is the preparation of food for the plant. Therefore, chloroplast is known as the kitchen of the cell.

27. Correct option: (A) Vacuoles

Vacuoles are membrane-bound organelles found in the cytoplasm of eukaryotic cells. They can vary in size, but they are typically large, bulged, and transparent structures that play various essential roles in the cell. One of their primary functions is to store water and maintain cell turgidity. Vacuoles can also store waste products, excretory substances, and even toxic materials in some cases.

28. Correct option: (D) a, b, and c

Chromosomes are present inside nucleus and are seen only during cell division and carries genes on it.

29. Correct option: (D) Cell Wall

Cell wall is a protective outermost layer, which is present surrounding the cell membrane. It is present in plant cells. It supports the contents of cell and protects from pathogen attack and wear and tear. It prevents water loss from plant cells. It gives shape to the plant. It is made up of cellulose and lignin.

30. Correct option: ** A-Thylakoid, B- Granum, C- Outer Membrane, D- Intermembranous space, E- Inner Membrane

Chloroplasts are found in all higher plants. It is oval or biconvex, found within the mesophyll of the plant cell. The size of the chloroplast usually varies between 4-6 μm in diameter and 1-3 μm in thickness. They are double-membrane organelle with the presence of outer, inner and intermembrane space. There are two distinct regions present inside a chloroplast known as the grana and stroma. Grana are made up of stacks of disc-shaped structures known as thylakoids or lamellae. The grana of the chloroplast consists of chlorophyll pigments and are the functional units of chloroplasts.

5. Reproduction in Animals

Solutions

LEVEL-I

1. Correct option: (C) Leopard

Animals that produce living offspring after the embryo has internally grown within the mother's body are said to be viviparous. The leopard in this instance is a viviparous animal. In the course of gestation, the foetus grows inside the mother's uterus in female leopards. The mother gives birth to live cubs once the foetus is fully formed. When compared to species that lay eggs or have less developed young, species like leopards benefit from this reproductive strategy because their young require a higher amount of protection and nurturing.

2. Correct option: (D) Cat

Mammals, including domestic cats, have epidermal hair, commonly referred to as fur or coat, which is a distinguishing characteristic. Cats have a thick layer of epidermal hair on their bodies, which the animals use for communication, camouflage, insulation, and other purposes. By trapping air near the skin, the hair aids cats in controlling their body temperature. In their native environments, it helps them blend in with their surroundings so they can stalk prey or avoid predators. They also receive sensory information about their surroundings through the tactile nature of cat fur, which improves their ability to navigate and interact with their environment.

3. Correct option: (D) All the above

Numerous creatures, including Hydra, algae, and fungi, have been found to reproduce asexually, which is a manner of reproduction devoid of gamete fusion. Simple aquatic organisms like the hydra reproduce asexually by budding. Photosynthetic organisms known as algae frequently reproduce asexually through fragmentation or spore production. Molds and yeasts are examples of fungi, which also use spore generation, budding, and fragmentation as asexual reproduction strategies. These species use asexual reproduction to rapidly produce young that are genetically identical or similar, allowing them to quickly adapt to their environment, fill new habitats, and continue to exist in different ecosystems.

4. Correct option: (B) Amoeba

Asexual reproduction known as binary fission occurs when a single parent cell divides into two identical daughter cells. A single-celled organism called an amoeba reproduces by binary fission. The amoeba's cytoplasm and nucleus divide during this phase, creating two daughter cells that are genetically identical to one another. Some unicellular creatures, like amoeba and bacteria, frequently reproduce in this way in order to proliferate and colonise their surroundings.

5. Correct option: (D) Stamen

The stamen, a flower's male reproductive organ, is absent from female flowers. The filament, which supports the anther, and the anther, which produces pollen grains, are the two primary components of the stamen. The pistil, which has three main components—the ovary, the style, and the stigma—is the reproductive structure in female flowers. Ovules are found in the ovary, where they undergo fertilisation to become seeds. The stigma is the receptive surface where pollen settles and germinates, leading to fertilisation, and the style is a tube-like structure that connects the stigma to the ovary.

6. Correct option: (B) Testes

The testes are one of the most important parts of the male reproductive system, which also comprises other organs. Sperm and male sex hormones like testosterone are produced by the testes. The epididymis, which matures and stores sperm, the vas deferens (sperm ducts), which transport sperm from the testes to the urethra, and accessory glands that generate fluids that combine with sperm to make semen are further parts of the male reproductive system. The testes are unique to the male reproductive anatomy, in contrast to the other alternatives (Fallopian tube, uterus, and ovary), which are components of the female reproductive system.

7. Correct option: (C) Uterus

The uterus, a crucial reproductive organ in all female mammals, including humans, is where the embryo grows. The zygote (fertilised egg) passes through numerous cell divisions after fertilisation to become a blastocyst. The blastocyst is subsequently implanted into the uterine lining, where it receives nourishment and assistance for growth. During pregnancy, the uterus offers a favourable environment for the embryo to transform into a foetus. This process results in the development of diverse embryonic structures as well as the growth of organs and systems, which ultimately results in the birth of a child that is fully developed.

8. Correct option: (A) Frog

Fertilization normally occurs externally in frogs. Male and female frogs both discharge sperm and eggs into the water during the breeding season. This external fertilisation takes place in watery habitats like ponds, lakes, or streams when the female frog's eggs and the male frog's sperm come into touch. Tadpoles are produced from the fertilised eggs, and these creatures eventually change into adult frogs. This process of fertilisation, which is an illustration of external fertilisation as opposed to internal fertilisation, in which fertilisation takes place inside the female's body, is ideally adapted to the frog's aquatic existence.

9. Correct option: (B) Tadpole

Tadpoles are the name for a frog's larval stage. Aquatic animals known as tadpoles are the offspring of frog eggs. As they grow into adult frogs, they often have a long tail, external gills for breathing underwater, and go through a transformation process. Tadpoles eventually lose their gills during metamorphosis, and their tails shorten and their legs begin to develop. They eventually change into froglets, which then grow into adult frogs that can survive on land. Tadpoles are an important stage in the life cycle of the frog, allowing them to colonise aquatic habitats before switching to a terrestrial way of life.

10. Correct option: (A) Male flower

The male reproductive organs of blooming plants are called pollen grains. They are created by the stamen and other male flower components. Anthers and filament make up the stamen. The pollen grain, which contains the male gametes, is created by the anther (sperm cells). When pollen grains are discharged from the anthers, they are frequently carried to female flowers by wind, insects, or other pollinators where they can fertilise the ovules and start the seed-forming process.

11. Correct option: (C) In Vitro Fertilisation

In vitro fertilisation is a technique used to create "test tube babies," commonly referred to as "IVF babies." An egg (ova) is fertilised by a sperm during IVF outside of the body, typically in a lab dish or culture medium. After fertilisation, the resultant embryos are given a few days to mature before being placed into the uterus in preparation for possible implantation and pregnancy. When natural conception is difficult or not possible, IVF is a reproductive technology used to treat infertility and help people or couples become pregnant.

12. Correct option: (C) 270–280 days

Human gestation, sometimes referred to as pregnancy, usually lasts between 270 and 280 days, or around 9 months. From the first day of the woman's last menstrual period until the baby is born, this period is measured. The precise duration of pregnancy can vary slightly from woman to woman and can be impacted by a number of variables, including the mother's and the baby's health, genetics, and environmental influences. However, the typical length is between 270 and 280 days.

13. Correct option: (D) Camel

The other choices, fish, frog, and lizard, are all vertebrate animals that fall within the classifications of Reptilia (for lizards) and Amphibia (for frogs) (for lizards). The physiological and anatomical properties of these animals are comparable. The camel, on the other hand, is a mammal and a member of the Mammalia phylum. All of the animals on the list are vertebrates, but the camel stands out from the other three due to its warm-bloodedness, the fact that it has mammary glands for producing milk, and its adaptations to desert settings. The camel differs from the other animals in that it is not aquatic, amphibious, or reptilian in nature.

14. Correct option: (D) Ostrich

The ostrich is an example of an oviparous animal, which are those that deposit eggs. African ostriches are enormous, flightless birds. They lay their eggs in ground nests, which are either incubated by the female ostrich or, in certain situations, also by the male. Prior to being laid, the eggs are internally fertilised, and they normally hatch after a predetermined incubation period. In order to secure the survival of their young in a variety of settings and conditions, oviparous animals, like the ostrich, have evolved this reproductive strategy. This technique frequently entails protective nests and systems to preserve the ideal environment for embryo development.

15. Correct option: (D) Reproduction

The fundamental life process that is essential to the survival of a species is reproduction. Through the development of children, it entails the creation of new individuals. In order to sustain genetic diversity and the survival of the species, reproduction makes sure that genetic information is transferred from one generation to the next. There are many different ways for different species to reproduce, including asexual reproduction, in which children are created from a single parent, and sexual reproduction, in which male and female gametes are fused to create offspring. A species' ability to reproduce and maintain its population over time would be compromised without it, which would eventually cause it to decrease and go extinct.

16. Correct option: (B) Ian Wilmut

An important historical milestone was reached when the first instance of cloning was carried out by pioneering British embryologist Ian Wilmut. He was an important contributor to the group at the Roslin Institute in Scotland that cloned a sheep dubbed "Dolly" successfully in 1996. Dolly was a significant accomplishment since she was the first mammal to be cloned utilising the somatic cell nuclear transfer method from an adult somatic cell. In addition to showing that it was possible to clone animals, Wilmut's discovery also provided new opportunities for genetic study, biotechnology, and the knowledge of cellular reprogramming. The study of biology and genetics advanced significantly as a result of this ground-breaking achievement.

17. Correct option: (C) Metamorphosis

A larval stage of an organism goes through considerable structural and physiological changes during the process of metamorphosis to become an adult form. This metamorphosis involves a number of developmental phases, many of which have unique external characteristics and ecological functions. The majority of insects, amphibians, and some aquatic species exhibit this behaviour. Examples of metamorphosis include the development of a tadpole into a frog or a caterpillar into a butterfly. In order to adjust to the new adult lifestyle and surroundings, it often entails the reorganisation and reabsorption of tissues as well as the restructuring of bodily parts.

18. Correct option: (A) Asexual

Asexual reproduction is defined as reproduction that occurs without the release of male and female gametes. Asexual reproduction produces children without the fusing of gametes, which are specialised reproductive cells, from two distinct parents. Since there is no genetic material mixing as occurs during sexual reproduction, this technique produces children that are genetically identical or highly similar. Binary fission, budding, and fragmentation are asexual reproduction techniques examples where the offspring receive the genetic features straight from a single parent.

19. Correct option: (D) Hermaphrodite and Bisexual animal

Hermaphrodites are animals with both male and female reproductive systems. As a result, they can reproduce with either males or females of their species since they have the capacity to generate both eggs and sperm. When referring to an animal that can reproduce in both sexes, the terms "bisexual animal" and "hermaphrodite" are frequently used interchangeably. As a result, D) Hermaphrodite and Bisexual Animal is the right response to the question because it includes both names and depicts creatures with both male and female reproductive systems.

20. Correct option: (B) Budding

In yeast cells, asexual reproduction occurs by budding. A tiny bud or protrusion on the parent cell emerges during budding, and as it expands and separates from the parent cell, it develops into a new organism. Under ideal circumstances, this mechanism enables yeast cells to multiply quickly and successfully. Yeast cells benefit from budding because it enables them to rapidly increase their population and adapt to changing conditions. Additionally, because the offspring receives a copy of the parent's genetic material, budding ensures genetic resemblance between the parent and offspring. Overall, yeast cells primarily reproduce asexually by budding, which aids in their survival and growth.

LEVEL – II

1. Correct option: (B) Reproduction

The biological process of reproduction is how living things create new members of the same species. It is a basic aspect of life that guarantees the survival of species over generations. It is possible for creatures to reproduce asexually and sexually, and this allows them to pass on their genetic makeup and characteristics to the following generation. The ability to make new people of the same sort is not immediately related to respiration, on the other hand, which is a metabolic activity involved in energy production and the exchange of gases.

2. Correct option: (A) 2

Asexual reproduction and sexual reproduction are the two basic types of reproduction. Asexual reproduction refers to the generation of genetically identical or extremely similar children from a single parent organism. Contrarily, sexual reproduction entails the union of gametes (sperm and eggs) from two distinct parents, resulting in genetic variation in the

progeny. These two methods of reproduction are important tactics used by different creatures to maintain their species.

3. Correct option: (A) Asexual

One parent usually participates in asexual reproduction, which is a sort of reproduction. Asexual reproduction is the process of producing offspring from a single parent organism without the joining of gametes or the use of genetic material from another person. Since there is no recombination of genetic material as occurs during sexual reproduction, this technique produces children who are genetically identical or highly similar. Numerous organisms have asexual reproduction, which is advantageous in circumstances that favour rapid population increase and colonisation.

4. Correct option: (B) Asexual reproduction

Amoebas are single-celled organisms that frequently reproduce asexually through a process known as binary fission. In binary fission, the amoeba's cell nucleus splits in half, then the remaining components of the cell divide as well. Due to this, two daughter cells are created that are genetically identical to one another and grow and develop into separate amoebas. Under the right circumstances, asexual reproduction enables amoebas to multiply quickly and colonise their surroundings.

5. Correct option: (C) Budding

The Cnidaria phylum member Hydra is a basic aquatic organism that reproduces asexually via a process known as budding. A new person, known as a bud, develops as a protrusion on the parent organism during this process. The bud gradually matures and separates from the parent to become a distinct and autonomous organism. Hydra frequently reproduce asexually by budding, which enables them to effectively increase their number under favourable circumstances.

6. Correct option: (D) Hydra

When an organism reproduces asexually, it creates offspring that are genetically identical to it. Simple creatures like bacteria and plants frequently reproduce in this way. Asexual reproduction is possible for the small aquatic organism Hydra, which is a member of the phylum Cnidaria. This process is known as budding. A new individual develops as an outgrowth or bud on the parent organism during budding. This bud develops, eventually separating to become a separate entity. Asexual reproduction through budding allows hydra to efficiently produce genetically identical offspring, promoting population growth in suitable environments.

7. Correct option: (B) Viviparous

The animals that directly reproduce young ones are called viviparous. Viviparity is a form of reproduction where the embryos grow inside the mother's body and are fed by the mother herself. The growing embryos in viviparous creatures, such as most mammals and certain reptiles, are given nutrition and oxygen through a placental link or other specialised structures, allowing them to grow and develop before giving birth as living offspring. Opposing this mode of reproduction is oviparity, in which animals lay their eggs outside of their bodies, and the embryos grow outside the mother's body. Therefore, the correct answer is B) Viviparous.

8. Correct option: (D) Sperm

Sperm makes up the male gamete. In many animal species, including humans, males generate sperm, which are specialised reproductive cells. Sperm cells are created with fertilisation in mind. They often have a tail (flagellum), which allows them to travel through fluid environments like the female reproductive canal to reach the egg. Sperm are essential

for sexual reproduction because their nuclei contain genetic information. The formation of a new person with a special blend of genetic features from both parents begins with the union of a sperm and a female gamete (egg), which creates a zygote. Therefore, sperm are necessary for a species to continue through sexual reproduction.

9. Correct option: (A) Sponges

Internal budding, commonly referred to as gemmulation, is an asexual reproduction process that is seen in sponges (phylum Porifera). Gemmules, which are tiny bud-like structures, develop within the parent sponge's body throughout this phase. The cells that make up these gemmules are grouped together and covered in a layer of defence. The parent sponge can discharge these gemmules into the nearby water when the environment becomes adverse, such as during periods of hard weather. The gemmules serve as a kind of survival capsule, shielding the sponge cells from harmful elements. When conditions are better, the gemmules can separate and grow into new sponge individuals, enabling sponges to endure and spread over a variety of aquatic settings. Internal budding in sponges is a remarkable adaptation that aids in their ability to reproduce asexually and withstand challenging ecological circumstances.

10. Correct option: (C) Fragmentation

Asexual reproduction known as fragmentation allows for the growth of a whole new organism from specific parts or cells of the original organism. The parent organism splits into various pieces throughout this process, and each piece has the ability to regenerate and grow into a brand-new, autonomous individual. Each fragment finally develops into a complete organism after the missing body parts are grown again by the pieces. Many different animals, including flatworms and some plant species, exhibit fragmentation, which enables them to procreate without the necessity for mating or the development of specific reproductive cells. The parent organism can produce several offspring using its own body parts thanks to this reproductive strategy, which increases population size.

11. Correct option: (B) Fragmentation.

Flatworms, also known as planarians, are capable of reproducing through a process called fragmentation. A flatworm can be divided into two or more parts via this technique of reproduction, and each portion can grow back into a whole creature. Each piece develops into a distinct individual and grows any missing body parts, such as the head and tail. Flatworms can produce new individuals from dismembered or separated body parts thanks to this type of asexual reproduction. While the animal kingdom also employs alternative asexual reproduction strategies, such as binary fission and budding, flatworms stand out for their use of fragmentation.

12. Correct option: (B) Sexual reproduction

An example of sexual reproduction occurs when male and female gametes from two distinct parents are fused. Genetic material is transferred between the parents during sexual reproduction, resulting in a genetically diverse progeny. This procedure typically calls for the development of specialised reproductive cells, or gametes, like sperm and eggs, which have half as many chromosomes as usual. These gametes join together during fertilisation to form a new person that combines genetic characteristics from both parents. Contrarily, asexual reproduction entails the generation of children without the requirement for gamete fusion, resulting in offspring that are genetically identical to their parents. B) Sexual reproduction is the right response, so.

13. Correct option: (C) Zygote

A sperm or egg (ova) cell is an example of a specialised reproductive cell known as a gamete. Gametes are necessary for sexual reproduction and are produced by both male and female

members of a species. A zygote is the structure formed when a male gamete (sperm) combines with a female gamete (egg). The first cell to form during fertilisation is called a zygote, and it combines genetic material from both parents. The embryo develops into a multicellular organism after undergoing fast cell divisions and development from the single-celled zygote. Thus, the development of sexual reproduction begins with the union of two gametes to produce a zygote.

14. Correct option: (B) Bisexual organism

An organism that has both male and female reproductive systems and is able to produce both male and female gametes (sperm and eggs) within a single individual is referred to as a hermaphrodite or a bisexual organism. With the use of one's own sperm and eggs, an individual may self-fertilize via this reproductive technique. Many animal species, particularly invertebrates like some fish, worms, and snails, are found to be hermaphrodites. The ability to reproduce even without a mate is ensured by this technique, however genetic variety may be lessened compared to cross-fertilization between different individuals.

Unisexual organisms, on the other hand, produce only one type of gamete. Male organisms produce sperm, and female organisms produce eggs. Some examples of unisexual organisms include humans, dogs, and cats.

15. Correct option: (C) Testes

As a component of the male reproductive system, the testes are where sperm, also known as spermatozoa, are created. The testes are two oval-shaped organs that hang outside of the male body in the scrotum. Seminiferous tubules, a large number of small tubules within the testes, are where spermatogenesis occurs. Sperms are created during the process known as spermatogenesis from unique cells known as spermatogonia. These cells go through a number of divisions and changes before becoming fully developed sperm cells. The mature sperm are released into the epididymis, a coiled tube attached to the testes, where they continue to develop and are kept until ejaculation, once they have fully developed. In the male reproductive system, the testes serve as the major location for the generation of sperm.

16. Correct option: (C) Sperm ducts

Sperm are transported from the testes to the penis by sperm ducts, commonly known as the vas deferens or ductus deferens. The male reproductive system includes these ducts, which act as a route for sperm to follow after ejaculation. The testes produce sperm, which mature as they pass through the testes' epididymis, a coil-like structure. In order to get to the urethra near the base of the penis, the sperm must first travel through the sperm ducts, which run from the epididymis through the pelvic cavity. The sperm are launched through the sperm ducts during ejaculation and combine with secretions from other accessory glands to form semen, which is then ejected from the penis. Thus, the sperm ducts play a crucial role in the transport of sperm for fertilization.

17. Correct option: (B) Four

A specialised cell used in sexual reproduction is a sperm. The head, the midpiece, the tail (sometimes referred to as the flagellum), and the neck are its four primary components. The genetic material is housed in the compact nucleus of the cranium. During fertilisation, the acrosome, a structure at the tip of the head, provides enzymes required for the sperm to pierce the egg's outer layers. There are many mitochondria in the midpiece, which supply the sperm with the energy it needs to migrate. The sperm can move through fluid environments thanks to the tail, or flagellum, which is a long, whip-like structure. The

midpiece and the head are connected by the neck. These four components work together to facilitate the sperm's journey toward the egg for the potential process of fertilization.

18. Correct option: (B) Female

The correct answer is B) Female. The flagellum, or tail, of a sperm is a whip-like feature that helps the sperm move forward in a fluid environment. The sperm's tail is principally in charge of enabling travel into the female reproductive canal in the context of human reproduction. The sperm must pass through the female reproductive system after being ejaculated during sexual activity in order to get to the fallopian tubes, where fertilisation may eventually take place. The sperm can travel via the cervical mucus, the uterus, and finally the fallopian tube where they may come into contact with an egg for fertilisation thanks to the movement of the tail. As a result, the tail's main job is to facilitate sperm passage inside the female reproductive system.

19. Correct option: (B) Penetration

A sperm cell's head contains a unique component called the acrosome. It is essential to the fertilisation process. When the sperm makes contact with the zona pellucida, the outer covering of the egg, enzymes from the acrosome, which are present, are released. These enzymes aid in removing the egg's defensive barriers, allowing the sperm to enter and unite with the egg cell. This penetration is a crucial stage in fertilisation because it enables the sperm's genetic material to unite with the egg's genetic material to create a zygote, which will later develop into an embryo. As a result, the acrosome's main purpose is to make it easier for the egg to penetrate.

20. Correct option: (A) Ova

The female reproductive system's egg-producing organs are the ovaries. A different name for the eggs is ova. One ovary is situated on either side of the uterus in the pelvic. The female gametes or eggs, known as ova, are produced by the ovaries, which are a component of the female reproductive system. These eggs, which contain the genetic material of the female, are necessary for sexual reproduction. Each of the countless ovarian follicles found in ovaries is home to an immature egg. Typically, one egg is chosen and matures within a follicle during the menstrual cycle before being released through the process of ovulation. Following the release of the egg and the subsequent fusing of a sperm and an egg, fertilisation can result in the creation of a foetus.

21. Correct option: (B) Ovulation

A developed egg, or ovum, is released from the ovary into the fallopian tube during ovulation, a biological process. The pelvis contains two tiny, almond-shaped structures called ovaries. They generate hormones and eggs. The ovarian follicle that contains the egg ruptures during ovulation, releasing the egg into the pelvic cavity. The fallopian tube, which is lined with small hair-like structures known as cilia and aids in capturing the egg there, then directs it toward the uterus. The sperm cell must fertilise the egg for it to embed in the uterine lining and begin a pregnancy. Ovulation often occurs 14 days or so before a woman's period begins.

22. Correct option: (C) Inside the female body

Male gametes (sperm) are placed inside the female body during internal fertilisation, where they fertilise the female gametes (eggs). Mammals, birds, and some reptiles frequently undergo this sort of fertilisation. In mammals, the male ejaculates sperm after inserting his penis into the female's vagina. The sperm next proceed to the ovaries, where they can fertilise the eggs, via the uterus and fallopian tubes. Compared to external fertilisation, internal fertilisation provides a number of benefits. The fact that the eggs are shielded from

predators and the elements is one benefit. The sperm are more likely to fertilise the eggs, which is an additional benefit.

23. Correct option: (A) Fertilization

The creation of a zygote occurs when a male gamete (sperm) combines with a female gamete (egg or ovum) during the process of fertilisation. This is the first step in a person's growth. A sperm cell enters the interior of the egg during fertilisation after breaking through its outer layers of protection. Once within, the sperm and egg nuclei combine to form a single cell, fusing their genetic material. The two parents' genetic material combines to form a unique set of DNA that determines the offspring's genetic features and characteristics. The crucial process of fertilisation takes place inside the female reproductive system, frequently in the fallopian tubes in humans.

24. Correct option: (B) Fallopian tube

Human fallopian tubes are where fertilisation occurs. The slender, tube-like fallopian tube, often referred to as the oviduct, joins the ovaries with the uterus. An egg is discharged from one of the ovaries during ovulation, and it travels through the fallopian tube. Fertilization can take place if both sperm are in the fallopian tube at the same moment. To get to the fallopian tube, the sperm must pass through the cervix and enter the uterus. The fertilised egg, also known as a zygote, travels via the fallopian tube after fertilisation in order to reach the uterus, where it implants and grows into a foetus. Therefore, human fertilisation takes place in the fallopian tube.

25. Correct option: (A) Fish

In fish, external fertilisation is seen. The male fish releases his sperm to externally fertilise the eggs while the female fish discharges her eggs into the water throughout this procedure. This sort of fertilisation takes place outside the female fish's body, typically in bodies of water like rivers, lakes, or oceans. Fish benefit from external fertilisation because it makes it possible to fertilise many eggs at once, increasing the likelihood of fertile offspring. Additionally, it gives sperm and eggs a chance to freely mix in the water, enhancing the genetic diversity of the offspring. However, external fertilisation also subjects the sperm and eggs to a variety of environmental elements, leaving them open to predators and other dangers. Overall, external fertilisation is a unique reproductive strategy observed specifically in fish

26. Correct option: (C) Implantation

The procedure through which the embryo fuses to the uterine wall is known as implantation. It is an important stage in the progression of a pregnancy. The endometrium, the uterus' lining, is where the embryo first penetrates. The embryo's trophoblast cells then infiltrate the endometrium and establish a blood vessel between it and the mother. This process, which enables the development of the placenta, a crucial organ created later in gestation, allows the embryo to obtain needed nutrients and oxygen from the mother's bloodstream, is crucial for the progression of pregnancy. In most cases, implanting happens 6 to 10 days after conception. After fertilisation, it can happen as soon as 4 days later or as late as 12 days later.

27. Correct option: (B) Foetus

The term "Fetus" refers to the developmental stage at which an embryo's individual bodily components may be recognised. The embryo has grown and differentiated significantly during this stage, and its main organs and structures have formed. Rapid cell division and the building of the framework for many organs and systems characterise the development

in the stage before this one, known as the embryo stage. The embryo's fundamental structures have formed during the foetal stage, which is a later stage of development.

28. Correct option: (B) Uterus

The uterus is not a part of the male reproductive organ. It is a female reproductive system that is essential to pregnancy. During pregnancy, the uterus serves as the foetus' home and source of nutrition. The reproductive system in males consists of the penis, testes, and seminal vesicles. Urine and semen are transported through the penis during sexual activity. The seminal vesicles help to create semen, whereas the testes produce sperm and male sex hormones. The only organ available that is not a part of the male reproductive system is the uterus.

29. Correct option: (D) Both B and C

Animals that give birth to live offspring are said to be viviparous. Animals that lay eggs include hens and silkworms, which are both oviparous. Young that can survive without their mother emerge from the eggs.

Eggs laid by hens develop into chicks. Within hours of hatching, the chicks are independent. Eggs laid by silkworms develop into caterpillars. The caterpillars undergo several moults before pupating to become moths. The moths have the ability to fly and lay eggs.

Humans are viviparous creatures. The fertilised egg grows into a foetus after implanting in the mother's uterus. Through the placenta, the mother feeds the foetus. After roughly nine months of gestation, the baby is delivered.

30. Correct option: (D) Hen

Hens do not undergo metamorphosis, a process that results in a significant shift in body composition. Hens grow and mature gradually without going through the specific metamorphic changes that these other organisms do, but animals like silkworms, fish, and butterflies go through metamorphic transformations during their life cycles.

Silkworms and butterflies emerge from their eggs as caterpillars, and as they develop, they undergo a series of moults. While the butterfly's last moult results in the adult butterfly, which has wings, the silkworm's final moult results in the adult silkworm, which is a moth. Fish grow through a series of moults after emerging from their eggs as larvae. The adult fish, which has scales and fins, is created after the last moult.

6. Reaching the Age of Adolescence

Solutions

LEVEL -I

1. Correct option: (B) 10–19 years

Between childhood and adulthood, adolescence is a period of profound physical, cognitive, and emotional change. It normally starts at age 10 and lasts until around age 18 or 19. People go through puberty during this time, which involves the development of secondary sexual traits and the capacity for reproduction. Adolescents also experience considerable brain development, which enhances their capacity for logic, judgement, and self-control. On an emotional level, adolescents might go through mood changes, increased self-consciousness, and the development of a personal identity. Given these considerations, the correct response to the question is B), which covers the age span in which the majority of teenagers go through these transformative changes.

2. Correct option: (C) Peers

Adolescents are at a developmental stage where they are seeking independence and creating their own identities, so it makes sense that they would be interested in spending time with their peers. Peers give teenagers a sense of community and acceptance, enabling them to explore their shared interests, morals, and worldviews. In addition to providing opportunities for social and emotional development, peer interactions help young people develop critical abilities including empathy, communication, and dispute resolution. Adolescents can share experiences and activities with their peers, which promotes a sense of togetherness and support for one another. In general, teenage social and emotional development depends heavily on peer relationships.

3. Correct option: (A) Larynx

Particularly in men, the Adam's apple is a noticeable feature of the neck. The thyroid cartilage, a component of the voice box or larynx, forms it. The larynx, which is found in the throat, is important for both creating sound and allowing air to reach the lungs. Due to the effects of testosterone throughout puberty, the laryngeal prominence, often known as the Adam's apple, is more pronounced in males. It helps to produce deep voices and acts as a protective structure for the vocal cords. Larynx is the proper response, so that is what it is. The alternate choices are false. The mouth is the aperture at the front, the pharynx is the throat, and the oesophagus is the food pipe.

4. Correct option: (B) Gametes

The reproductive organs create gametes, which are reproductive cells. The testes are the male reproductive system. The male gametes, or sperm, are created in the testes. The ovaries are a female's reproductive organs. The female gametes, or eggs, are created in the ovaries. A gamete is required for reproduction. A zygote, which is an egg that has been fertilized, is created when a sperm and an egg join. The zygote can then grow into a brand-new person.
The alternate choices are false. Chemicals known as hormones are created by the endocrine system. The male reproductive cells of plants are called pollen grains.

5. Correct option: (D) 21 years

The legal age for female marriage is 21. This means that a girl must be at least 21 years old in order to lawfully get married. In order to ensure that people are capable of making such an important decision and completely comprehend the obligations and repercussions that come with marriage, the legislation has imposed this age restriction. It also aims to safeguard young women from being pressured into marriages at an early

age, giving them more time to pursue their education, job goals, and personal development before committing to a committed life partner.

6. Correct option: (C) Endocrine gland

The answer to the question is C) Endocrine gland. Hormones are produced and released into the bloodstream by endocrine glands. These hormones are essential for the growth and control of the human reproductive system. They manage the development and maturation of reproductive organs such the ovaries and testes, as well as the gamete production (sperm and eggs). Endocrine glands secrete hormones like oestrogen and testosterone, which are crucial for the maturation of secondary sexual traits, the onset of puberty, and the maintenance of reproductive capabilities into adulthood. Endocrine glands are therefore essential for the growth and operation of the human reproductive system.

7. Correct option: (D) All of the above

Digestion, respiration, and growth are a few of the different processes that make up an organism's character. The term "digestion" describes the process by which food is broken down into smaller molecules that the body can absorb and use. Gases like oxygen and carbon dioxide are exchanged during respiration to create energy. The term "growth" describes how an organism's size or cell count increases. The existence and operation of living things depends on each of these processes. Since all of these processes contribute to the general nature of living things, the correct response is D) All of the above.

8. Correct option: (C) Pituitary

Emotional regulation is the responsibility of the pituitary gland. Due to its ability to create and release hormones that control a variety of biological processes, including emotions, it is frequently referred to as the "master gland." At the base of the brain, the pituitary gland connects to the hypothalamus. It releases hormones that have an impact on stress levels, mood, and general mental health. C) Pituitary is the right response to the question.

The alternate choices are false. The male gonads are the testes, while the female gonads are the ovaries. Although they create sex hormones, they do not have emotional control. A hormone called adrenaline is released in response to stress, yet it has little effect on mood.

9. Correct option: (C) Adolescence

The period between childhood and adulthood is known as adolescence. People go through major physical, emotional, and cognitive changes throughout this transitory stage. Teenagers go through puberty during this time, which entails the emergence of secondary sexual traits. Additionally, they go through heightened emotions, greater independence, and an identity crisis. As people start to create their own opinions, values, and social networks, adolescence is a key time for social and emotional growth. It's a period of exploration, self-discovery, and getting ready for the obligations and difficulties of maturity. Adolescence is the proper response, which is C.

10. Correct option: (B) Maximum

Children go through a considerable growth spurt during adolescence, which results in an increase in height. Numerous elements, including nutrition, hormones, and heredity, have an impact on this growth spurt. It's common to refer to a child's height during this time as their maximum height. This is due to the fact that growth slows and eventually stops after adolescence, resulting in the establishment of their adult height. As a result, B) Maximum is the correct response to the question because it captures the maximum height that children can reach during adolescence.

11. Correct option: (A) 25%

The percentage of smokers in India is A) 25%, in response to the question. This indicates that in India, there are about 25 percent of smokers. The fact that such a high percentage of the population smokes, suggesting that smoking is a widespread problem in the nation,

should not be overlooked. The high rate of smoking emphasizes the need for strong policies to discourage smoking, advance public health, and run awareness efforts to lessen the negative impacts of smoking on people and society at large.

12. Correct option: (A) Adam's apple

The Adam's apple, a bulge of cartilage on the front of the neck, is more noticeable in men than in women. It is brought on by the thyroid cartilage, one of the cartilages that make up the larynx, growing. Another name for the larynx is the voice box. Sound is produced by it, which is situated in the neck. To make sound, the two tissue bands that span the larynx called the vocal cords vibrate. There is no sound coming from the Adam's apple. But when the thyroid cartilage grows, it may enlarge and become more noticeable, giving the voice a deeper tone.

The other options are incorrect. The voice box is the complete larynx, and the voice chamber is a different name for the larynx. Thyroid is a gland that is located in the neck, but it is not part of the larynx.

13. Correct option: (B) Sebaceous glands

Sebaceous glands are another name for oil glands. These glands are in charge of creating sebum, an oily material that is vital for lubricating the skin and hair. Sebum aids in retaining moisture in the skin and preventing drying out. Additionally, it aids in defending the skin from aggressors including bacteria and toxins from the environment. Except for the palms of the hands and the soles of the feet, the body is covered in sebaceous glands. On the face, scalp, and upper back, they are most prevalent. Oily skin and acne can result from these glands being hyperactive.

The alternate choices are false. Lacrimal glands create tears, sweat glands create perspiration, and serumenous glands create earwax.

14. Correct option: (C) Menarche

The right response is C) Menarche. The term "menarche" refers to a girl's first period, which normally begins when she is between the ages of 12 and 14. It signals the start of female puberty and the start of the reproductive stage. Other puberty symptoms, like breast development, pubic hair growth, and broadening of the hips, are frequently present when a woman experiences her first menstrual period. There may be some discomfort during menarche, such as cramping and bloating.

Contrarily, menopause is the term used to describe the end of women's menstruation and fertility, which typically happens between the ages of 45 and 55. A freshwater ecosystem's succession of plant species is referred to as a "hydrarch" in ecology.

15. Correct option: (C) 21

For men, 21 is the minimum age to get married. This means that in order for a man to lawfully engage into a marriage contract, he must be at least 21 years old. In order to ensure that people have attained a particular degree of maturity and are capable of making thoughtful decisions on their marriage, the law has imposed this age requirement. Men are said to be emotionally and mentally ready for the obligations and commitments that come with marriage by the age of 21, having adequate life experience. This age restriction also guards against the possibility of exploiting or pressuring younger people into marriages before they are ready.

16. Correct option: (C) Iron

Because iron is needed for the creation of hemoglobin, the protein that carries oxygen to the body's numerous tissues and organs, red blood cells (RBCs) are produced as a result. Iron deficiency anemia is a disorder that develops when the body is unable to create enough hemoglobin due to a lack of iron in the body. Shortness of breath, weakness, and

exhaustion are some of the signs of this illness, which leads to decreased RBC production. Iron is therefore essential for both the creation of RBCs and the body's general ability to transport oxygen.
The alternate choices are false. It is not necessary for the creation of RBCs to have magnesium. Additionally, neither boron nor zinc are required for the creation of RBCs, but they do have a role in other physiological processes.

17. Correct option: (B) Adolescents

Secondary sexual characters are seen in adolescents. During adolescence, individuals undergo significant physical changes as they transition from childhood to adulthood. These changes are driven by the activation of sex hormones, such as testosterone and estrogen, which lead to the development of secondary sexual characteristics. These characteristics include the growth of facial and body hair in males, breast development in females, deepening of the voice in males, and widening of the hips in females. These changes are essential for reproductive purposes and help distinguish individuals as sexually mature adults.

LEVEL -II

1. Correct option: (C) 19

The period between childhood and adulthood known as adolescence is characterized by various physical, emotional, and social changes. It is generally accepted to start around the time of puberty, which typically occurs when a child is 11 years old, and last until they are in their late teens or early twenties. Although there may be some variance in the precise age range, 19 is typically regarded as the upper age limit of adolescence. People are often on the verge of adulthood at this age, with many already taking on adult duties and making significant life decisions.
Adolescence brings about a variety of physical, mental, and emotional changes that can be difficult for both the teenager and their parents to deal with. The role of parents during this time should be one of understanding and support. They can also help their adolescent by providing them with information about puberty, healthy relationships, and decision-making.

2. Correct option: (B) Hormones

Chemicals called hormones are injected right into the bloodstream. They are created by a number of bodily glands, including the pituitary, thyroid, and adrenal glands. As messengers, hormones transmit signals from one area of the body to another. They control a variety of vital bodily processes, such as metabolism, reproduction, mood, and growth and development. Hormones go to their intended tissues or organs after being released into the bloodstream, where they attach to particular receptors and start a reaction. Hormones may connect with various bodily parts rapidly and effectively because to this direct release into the bloodstream, maintaining correct coordination and functioning.
The other options are incorrect. Blood is not a chemical substance, and lymph fluid is not released directly into the bloodstream.

3. Correct option: (B) Hormones

Glands create hormones in the body. The body's glands are specialized organs that release hormones into the blood. These hormones serve as chemical messengers and are essential for controlling a number of body processes and functions. They aid in the regulation of many physiological processes, including growth, metabolism, reproduction,

mood, and many more. The body couldn't sustain homeostasis and function normally without hormones.

A substance that travels throughout the body is called blood. Red blood cells, white blood cells, platelets, and plasma make up its composition. A transparent fluid called lymph fluid circulates throughout the lymphatic system. It aids in infection prevention and waste product removal from the body.

4. Correct option: (A) Endocrine

Endocrine is the correct answer. Since endocrine glands lack ducts or tubes to carry their secretions, they are also referred to as ductless glands. Instead, they immediately release hormones into the circulation. These hormones are subsequently transported to the body's target cells or organs, where they control a variety of physiological functions and uphold homeostasis. The pituitary gland, thyroid, adrenal, and pancreatic glands are a few examples of endocrine glands. These glands are vital for controlling the body's vital processes as growth, metabolism, reproduction, and others. Contrarily, exocrine glands do contain ducts that direct their secretions to a particular place. Exocrine glands include, for instance, salivary and sweat glands.

5. Correct option: (D) Brain

The pituitary gland is located at the base of the brain. It is a tiny, pea-sized gland that is frequently called the "master gland" since it is so important in controlling several organ functions. The pituitary stalk, a stalk-like structure, connects the pituitary gland to the hypothalamus, a brain region. Hormones that regulate development, metabolism, reproduction, and other critical bodily functions are created and released by this gland. It can receive signals from the hypothalamus thanks to its position at the base of the brain, and it can react by releasing the appropriate hormones into the bloodstream.

6. Correct option: (A) pituitary gland

At the base of the brain, there lies a pea-sized gland called the pituitary. It generates hormones that regulate the activity of other glands, such as the thyroid gland, adrenal glands, and ovaries, which is why it is known as the "master gland." Growth hormone is one of the hormones generated by the pituitary gland (GH). GH is in charge of promoting the development of bones and muscles. It also affects the growth of other tissues, including the liver, lungs, and heart. Growth hormone is not secreted by the pancreas, testes, or any of the mentioned organs.

7. Correct option: (C) Goitre

Goitre is caused by a thyroxine deficit. The thyroid gland produces the hormone thyroxine, which is essential for controlling metabolism. Thyroxine shortage causes the thyroid gland to expand in an effort to make up for the reduced hormone output. Goitre is the medical term for this thyroid gland hypertrophy. Goitre can result in a neck swelling that is apparent and create symptoms including breathing or swallowing problems. C. Goitre is the appropriate response to the question.

8. Correct option: (A) Kidneys

The small, triangular-shaped adrenal glands are found on top of each kidney. They are a vital component of the endocrine system and produce and release hormones like cortisol, aldosterone, and adrenaline. The adrenal glands' location above the kidneys gives them direct access to the bloodstream, which facilitates the speedy and effective transportation of the hormones they manufacture throughout the body. Due to their near proximity, the kidneys and adrenal glands can communicate with one another and coordinate their efforts to control many biological processes like blood pressure, metabolism, and stress response.

9. Correct option: (C) Oestrogen and Progesterone

The female sex hormones, notably oestrogen and progesterone, are produced by the ovaries in females. Oestrogen is essential for the growth and control of the female reproductive system, as well as for the development of the breasts and the menstrual cycle. Contrarily, progesterone plays a role in getting the uterus ready for pregnancy and keeping a healthy pregnancy. Together, these hormones control ovulation, regulate the menstrual cycle, and aid in the growth and upkeep of the female reproductive system.

10. Correct option: (A) Testes

The male sex hormone testosterone is created by the testes, which are the male reproductive organs. These scrotal glands are essential for the growth of male reproductive characteristics and abilities. The expansion of the male reproductive organs and the development of secondary sexual features including face and body hair, a deeper voice, and these other sexual traits all depend on testosterone. Along with controlling sperm generation, muscle mass, bone density, and sexual health in general, it also controls a number of physiological functions.

Options B (Ovary) and C (Penis) are erroneous because they have nothing to do with testosterone production. Given that males' testicles do indeed create testosterone, Option D (None) is equally erroneous.

11. Correct option: (B) Sperm

Testosterone, the primary male sex hormone, plays a pivotal role in the development and maintenance of male reproductive traits and functions. It stimulates the production of sperm cells in the testes through a process called spermatogenesis. Sperm cells are crucial for male fertility and are responsible for carrying genetic information to fertilize an egg cell during sexual reproduction. While testosterone has other physiological effects in the body as well, its role in supporting the production of sperm cells is specifically significant in the context of male reproductive health.

Options A and C (Ova and Both A and B) are incorrect as they pertain to female reproductive components and do not align with testosterone's role in sperm production. Option D (None) is also incorrect given the established connection between testosterone and sperm cell development.

12. Correct option: (B) Pancreas

An essential organ in the human body that is necessary in controlling blood sugar levels is the pancreas. It secretes the hormones insulin and glucagon, which cooperate to regulate the metabolism of glucose. Glucagon boosts blood sugar levels by encouraging the release of glucose from the liver, whereas insulin helps lower blood sugar levels by enabling the uptake of glucose into cells. In order to regulate blood sugar levels and preserve overall health, there must be a dynamic interaction between the hormone’s insulin and glucagon.

The other options (A. Testes, C. Ova, D. Sperm) are not relevant to the secretion of insulin and glucagon.

13. Correct option: (C) Diabetes

The chronic medical illness known as diabetes is characterized by the body's inability to generate or use insulin, a hormone essential for controlling blood sugar levels. Type 1 diabetes and type 2 diabetes are the two most common kinds of the disease. In people with type 1 diabetes, the immune system of the body assaults and kills the cells that make insulin in the pancreas, which results in a lack of insulin. In type 2 diabetes, the body either stops producing enough insulin to meet the body's needs or develops a resistance to the effects of insulin. As a result, blood sugar levels increase, leading to a number of health issues. Maintaining blood sugar levels within a healthy range requires using a combination of food, exercise, medication, and, when necessary, insulin therapy.

14. Correct option: (A) Secondary sexual characters.

Between childhood and adulthood, adolescence is a period of profound physical, psychological, and emotional change. Secondary sexual traits start to show throughout this stage as a result of hormonal changes. These traits, which are particular to each gender and play a role in sexual maturation and differentiation rather than being directly related to reproduction, are gender-specific. Male secondary sexual characteristics can manifest as voice deepening, a rise in face and body hair, an increase in muscular mass, and alterations in body composition. These changes in females include the expansion of the breasts, broadening of the hips, and pubic and underarm hair growth. These traits set males and females apart, and they also contribute to attraction and reproduction, but they are different from primary sexual traits, which concern the reproductive organs directly.

15. Correct option: (D) All the above

Physical alterations brought on by hormonal changes during puberty are known as secondary sexual features. Males and females can be distinguished from one another based on these changes, which are suggestive of sexual development but are not directly related to the reproductive organs themselves. The voice alters as the larynx (voice box) develops, giving men a deeper voice. The growth spurt that comes with puberty, when bones lengthen, is what causes height increases. Both men and women can grow pubic hair, also known as genital hair, however the patterns and rates of growth might vary. These traits contribute to the total physical alteration, along with other modifications like the development of breasts in women and broader shoulders in men.

16. Correct option: (B) Menarche

Menarche, which normally occurs between the ages of 10 and 14, is the first time an adolescent girl has menstruation. The beginning of the female reproductive years is a crucial milestone. A girl's body experiences hormonal changes during menarche that cause the reproductive system to mature, including the development of the uterus and ovaries. When a girl's menstrual cycle starts, her body is ready to start releasing eggs and prepare for a possible pregnancy. Menarche varies from person to person and is affected by things like genetics, diet, and general health. It's a significant occasion in a young woman's life because it represents her entry into adulthood and the possibility of becoming pregnant.

17. Correct option: (A) Menopause

The menopause is a normal biological process that signals the end of a woman's reproductive years. It happens when a woman's ovaries stop producing eggs, which results in a permanent end to her monthly cycles. Generally speaking, this occurs between the ages of 45 and 55, however it can happen at a different time for each person. Hormonal changes, particularly a drop in oestrogen levels, characterise menopause. Women consequently go through a variety of physical and psychological changes, including mood swings, hot flashes, and changes in bone density. Menopause marks the end of women's reproductive potential and the start of a new stage of life in which they are no longer able to conceive naturally. It is distinct from menarche, which denotes the start of the reproductive cycle and involves the first onset of menstruation.

18. Correct option: (C) 23 pairs

The average human cell has 46 chromosomes, which are divided into 23 pairs. Each pair of chromosomes has one from each parent, with one member coming from the mother and the other from the father. These chromosomes include genetic material that affects a variety of features and qualities. The final pair of the 23 pairs consists of sex

chromosomes that determine an individual's biological sex, while the other 22 pairs of the 23 pairs are autosomes responsible for features unrelated to gender. The response "23 pairs" emphasises the significance of these genetic structures in our growth and functioning while also precisely reflecting the number of chromosomes present in a human cell.

18. Correct option: (C) 2

Two sex chromosomes are present in both males and females. The cells of humans contain a total of 46 chromosomes, which are divided into 23 pairs. The majority of qualities are inherited through the use of 22 pairs of these, which are referred to as autosomes. The sex chromosomes in the remaining pair establish a person's biological sex. Males have one X and one Y chromosome, while females have two X chromosomes (XX) (XY). Each person only possesses two sex chromosomes, which are essential for determining their reproductive and sexual traits, regardless of the combination (XX or XY). This response highlights the fact that everyone, regardless of gender, possesses two sex chromosomes.

19. Correct option: (C) Boy, Girl

In humans, the presence of particular sex chromosomes affects how sex is determined. The Y chromosome is inherited from the father, making a child biologically male, whereas the X chromosome is inherited from the mother, making a child biologically female. This is due to the fact that although females have two X chromosomes, males only have one X and one Y chromosome (XY) (XX). The genes that control male development, including the growth of male reproductive organs, are found on the Y chromosome. On the other hand, female development results from the absence of the Y chromosome. As a result, a child who inherits the Y chromosome will grow up to be a boy and a child who inherits the X chromosome will be a girl.

20. Correct option: (A) Proteins

Proteins are necessary for bodybuilding, tissue repair, and maintenance. Amino acids, the building blocks of life, are found in proteins. These amino acids are essential for many biological functions, including the development, maintenance, and repair of tissues like muscles, organs, skin, and bones. Proteins are crucial for muscle growth and recovery after exercise in the context of bodybuilding and fitness. Muscle tissues may micro-tear during physical activity, which needs to be mended in order for muscles to grow. Consuming protein gives the body the essential amino acids to help repair and rebuild muscle fibers, which over time results in more muscular mass and strength. Additionally, proteins serve a variety of other essential bodily roles, such as enzyme

21. Correct option: (D) All the Above

AIDS (acquired immunodeficiency syndrome) is contracted by coming into contact with an infected person through a variety of channels. This covers sexual intercourse, sharing contaminated needles or syringes, and transfusions of blood. The AIDS virus can be spread through blood transfusions using infected blood. The chance of the virus spreading from one person to another when intravenous drug users share needles is very high. Sexual intercourse, particularly unprotected sex with an infected partner, is one of the most popular means of transmission, nevertheless. Human Immunodeficiency Virus (HIV), which causes AIDS, can be found in vaginal secretions, blood, and breast milk. Sexual activity without protection with an infected person so increases the risk of transmission. Therefore, the answer "All the Above" accurately reflects the multiple routes through which AIDS can spread, emphasizing the importance of awareness and prevention measures.

22. Correct option: (B) 28

Ovulation, or the periodic release of eggs from the ovary, occurs in women of reproductive age typically every 28 days. A developed egg is produced from one of the ovaries during ovulation, and it travels down the fallopian tube before potentially being available for fertilization by a sperm cell. Ovulation is a significant occurrence in the menstrual cycle. The average cycle length is 28, though there can be variances depending on the person. Hormonal changes, especially involving the interaction of the hypothalamus, pituitary gland, and ovaries, control this process. Ovulation is a crucial stage in a woman's reproductive cycle since it is when her chances of conceiving are at their maximum.

23. Correct option: (A) Zygote

During fertilization, a zygote is created. A male gamete (sperm) and a female gamete (ova or egg) combine during fertilization to create a zygote. The first cell of a new organism, known as a zygote, carries the genetic makeup of both parents united. The formation of a new person begins with this merger of genetic material. The zygote begins to generate the embryonic features that will eventually mature into a mature organism as it goes through cell division. All of the genetic instructions required to direct the development of numerous tissues, organs, and systems as the embryo grows and matures are present in the zygote.

24. Correct option: (C) Month

It is true that the Latin word "mensis," which meaning "month," is where the word "menstrual" comes from. The phrase is used to describe the cycle of alterations that take place in the female reproductive system on a monthly basis, including the removal of the uterine lining during menstruation. Generally speaking, the menstrual cycle lasts for about 28 days, though there can be variances. The term "menstrual" reflects the cyclical nature of this process, which occurs roughly once a month.

25. Correct option: (C) Sweat and sebaceous

Puberty causes the sebaceous and sweat glands to become increasingly active, which increases perspiration. Significant hormonal changes, principally brought on by the maturing of the reproductive system, take place throughout the puberty stage of development. The sweat glands, often referred to as eccrine glands, are essential for controlling body temperature because they secrete perspiration that evaporates from the skin's surface and cools the body. Sebaceous gland activity also increases during puberty. Sebum, an oily substance produced by these glands, aids in lubricating and protecting the skin and hair. Changes in body odor, acne, and other skin disorders frequently linked to puberty-related hormonal changes might be brought on by the increased activity of the sweat and sebaceous glands.

7. Pollution of Air and Water (Natural Resources)

Solutions

LEVEL-I

1. Correct option: (C) 6 months

Every PUC certificate is valid for a period of 6 months. Therefore, every 6 months, you need to get your car checked and re–tested for pollution emission levels to get a fresh PUC certificate issued.

2. Correct option: (A) 21%

Earth's atmosphere is composed of about 78 percent nitrogen, 21 percent oxygen, 0.9 percent argon, and 0.1 percent other gases. Trace amounts of carbon dioxide, methane, water vapor, and neon are some of the other gases that make up the remaining 0.1 percent.

3. Correct option: (C) Carbon dioxide

In the decomposition process, different products are released: carbon dioxide (CO_2), energy, water, plant nutrients and resynthesized organic carbon compounds.

4. Correct option: (C) Visakhapatnam

NTPC Simhadri Super thermal power plant at Parawada, Visakhapatnam, Andhra Pradesh.

5. Correct option: (C) Nuclear power

Nuclear energy originates from the splitting of uranium atoms – a process called fission. This generates heat to produce steam, which is used by a turbine generator to generate electricity. Because nuclear power plants do not burn fuel, they do not produce greenhouse gas emissions.

6. Correct option: (D) 1984

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

7. Correct option: (C) Chemicals

Industrial waste exhaust gases contain large amounts of toxic sulfurized gases such as SO_2, H_2S, CS_2, and sulphur-containing organic compounds. Removal of these gases from exhaust gases is one of the major applications of activated carbons.

8. Correct option: (D) All

Thermal power plant emissions cause asthma, Chronic Obstructive Pulmonary Disease Stunts lung growth, lung cancer, cardiac disease and congestive heart failure

9. Correct option: (A) PAN

Peroxyacetyl nitrate (PAN) is a secondary pollutant because it is formed in the atmosphere after the emission of primary pollutants which is a part of smog that causes air pollution.

10. Correct option: (B) Yamuna

The Taj Mahal was built on the banks of the river Yamuna, between 1631 and 1643, by the Mughal emperor Shah Jahan (r. 1628-58) as a mausoleum for his beloved wife Arjumand Banu Begum, who was also known by the title Mumtaz Mahal.

11. Correct option: (C) Mercury

The source and transmission mode are contaminated fish and shellfish. The etiologic agent is methylmercury, a by-product of acetaldehyde production, which was discharged from the Chisso factory from 1932 until 1968.

12. Correct option: (A) 31%

Forests cover 31 percent of the world's land surface, just over 4 billion hectares. (One hectare = 2.47 acres.) This is down from the pre-industrial area of 5.9 billion hectares.

13. Correct option: (D) All the above

Refrigerators, air conditioners, different types of sprays, fire extinguishers, and paints released chlorofluorocarbons in the air. And this emission contributed to the depletion of the Ozone layer.

14. Correct option: (B) July

Van Mahothsav or Vanamahothsava, lit. 'Forest festival', is an annual one-week tree-planting festival in India which is celebrated in the first week of July.

15. Correct option: (D) CNG

Petrol, diesel, and kerosene are petroleum products and they produce pollutants like carbon-dioxide and smoke. CNG or compressed natural gas is a smoke-free gas and does not spread pollution, and thus used in our vehicles. Therefore, CNG is considered as an eco-friendly fuel.

16. Correct option: (A) Nitrates and phosphates

Addition of phosphates and nitrates/fertilizers into water leads to nutrient enrichment which is also called as eutrophication. It is the process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen. It results in the loss of biodiversity.

17. Correct option: (A) CFCs

Gaseous CFCs can deplete the ozone layer when they slowly rise into the stratosphere, are broken down by strong ultraviolet radiation, release chlorine atoms, and then react with ozone molecules.

18. Correct option: (B) CO2 and SO2

Acid Rain is slightly acidic because it contains dissolved carbon dioxide (CO2). Sulphur dioxide (SO2) and Nitrogen oxides (NOx) which are normally present in the air.

19. Correct option: (A) Electrostatic precipitator (ESP)

Electrostatic precipitators are installed to control the emission of Suspended Particulate Matter (SPM). These precipitators can remove over 99 percent of the particulate matter present in the exhaust from a thermal power plant.

20. Correct option: (C) Methyl Isocyanate

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

LEVEL-II

1. Correct option: (D) All of these

Instead of the fossil fuels for our energy requirements we can use Solar Energy, Hydropower and Wind energy as there is least likelihood of release of greenhouse gases into the atmosphere.

2. Correct option: (C) 21%

Earth's atmosphere is composed of about 78 percent nitrogen, 21 percent oxygen, 0.9 percent argon, and 0.1 percent other gases. Trace amounts of carbon dioxide, methane, water vapor, and neon are some of the other gases that make up the remaining 0.1 percent.

3. Correct option: (B) Nitrogen

Sulphur dioxide, carbon dioxide, and methane are directly linked with global warming, as these gases increase heat within the aerial blanket surrounding the earth. But nitrogen is not responsible for global warming, as it is 78% in the atmosphere with no heat trapping history.

4. Correct option: (A) Infrared radiations

Molecules of carbon dioxide (CO2) can absorb energy from infrared (IR) radiation. The energy from the photon causes the CO2 molecule to vibrate.

5. Correct option: (A) Replace

The 3R's mantra that is used for the conservation of water includes reduce, reuse and recycle. Thus, replace is not a way to conserve water.

6. Correct option: (C) DDT

DDT and its metabolites are readily adsorbed onto sediments and soils, which can act both as sinks and as long-term sources of exposure. Because of its strong tendency to be adsorbed onto surfaces, most DDT that enters water is and remains firmly attached to soil particles.

7. Correct option: (B) Distillation

Distillation procedures will give us water free from all impurities. The distillation process removes almost all impurities from water. Distillers are commonly used for removing nitrate, bacteria, sodium, hardness (dissolved solids), most organic compounds, heavy metals and radionuclides from water.

8. Correct option: (C) Drip irrigation

Drip Irrigation system helps water to reach roots effectively. The most suitable method of irrigation for areas having low rainfall and strong winds is drip irrigation, as water will be suitably used in drip irrigation with high efficiency in low rainfall and strong wind areas.

9. Correct option: (D) Chlorination

Chlorination is a commonly used chemical method for purifying water. It kills germs in the water.

10. Correct option: (A) Air Pollution

Sulphur dioxide and nitrogen dioxide are pollutants emitted into the atmosphere as a result of the combustion of fossil fuels such as diesel and coal. Sulphur dioxide, NO2 emissions, and primarily carbon-based particles have corroded and damaged the Taj's beautiful white exterior, leaving it with a yellow sheen.

11. Correct option: (A) Nitrogen dioxide

In the lung, nitrogen dioxide (NO2) hydrolyzes to nitrous acid (HNO2) and nitric acid (HNO3), which can then cause chemical pneumonitis and pulmonary edema. Because NO2 is poorly water soluble, it hydrolyzes more slowly than other water-soluble gases, resulting in deep lung injury in the bronchioles and alveoli.

12. Correct option: (B) CFCs

Many refrigerants, such as chlorofluorocarbons (CFCs) damage the ozone layer, while others are extremely potent greenhouse gases. In fact, one kilogram of the refrigerant R410a has the same greenhouse impact as two tonnes of carbon dioxide, which is the equivalent of running your car for six months.

13. Correct option: (C) The sun’s UV rays

The stratospheric ozone layer is Earth's “sunscreen” – protecting living things from too much ultraviolet radiation from the sun.

14. Correct option: (A, B, C) CO2, CO, CH4

CO2 (Carbon dioxide) is an air pollutant, primarily due to its role in contributing to climate change when present in excessive amounts. CO (Carbon monoxide) is an air pollutant that can be harmful when present in high concentrations, as it can interfere with the body's ability to transport oxygen in the blood. CH4 (Methane) is another air pollutant that contributes to climate change when released into the atmosphere, as it is a potent greenhouse gas.

15. Correct option: (A) Swimming in the pond

Swimming in the pond will generate the least water pollution as no harmful chemicals or pollutants would get released in the process.

16. Correct option: (B) Leakage of poisonous gas

On December 3 1984, more than 40 tons of methyl isocyanate gas leaked from a pesticide plant in Bhopal, India, immediately killing at least 3,800 people and causing significant morbidity and premature death for many thousands more.

17. Correct option: (C) Water Pollution

The addition of harmful materials to water that reduces its quality is called water pollution.

18. Correct option: (C) Reduced oxygen carrying capacity of RBCs

This is because when carbon monoxide and oxygen both are available to RBCs then the RBCs will bind 200 times faster with carbon monoxide than oxygen, thus forming carboxyhaemoglobin rather than oxyhaemoglobin. This carboxyhaemoglobin reduces the oxygen carrying capacity of RBCs (since RBCs contain haemoglobin, a red-coloured protein).

19. Correct option: (D) Large amount of salt

Seawater contains a large amount of salt in it. Drinking a large amount of seawater makes a person dehydrated. Hence, it is not good for health.

20. Correct option: (B) Reduce Ganga water pollution

The Ganga Action Plan or GAP was a program launched by Rajiv Gandhi in April 1986 to reduce the pollution load on the river. But the efforts to decrease the pollution level in the river became more after spending Rs 9017 million.

21. Correct option: (C) Carbon monoxide

The Ganga Action Plan or GAP was a program launched by Rajiv Gandhi in April 1986 to reduce the pollution load on the river. But the efforts to decrease the pollution level in the river became more after spending Rs 9017 million.

22. Correct option: (D) All

They are used in the manufacture of aerosol sprays, blowing agents for foams and packing materials, as solvents, ACs and as refrigerants in Refrigerators.

23. Correct option: (B) Acid Rain

Pollutants in the air, such as Sulphur dioxide and Nitrogen dioxide, react with water vapour in the atmosphere to produce Sulphuric acid and Nitric acid, respectively.

These acids react with rain, causing it to become acidic.

Marble cancer is a phenomenon caused by acid rain corroding building materials like marble.

24. Correct option: (A) Spring Water

High-quality spring water is rich in magnesium, potassium, calcium, sodium, and other trace minerals. It is naturally more alkaline than tap water, which can create an environment in the human body less vulnerable to disease proliferation.

25. Correct option: (C) Reduced oxygen carrying capacity of RBCs

This is because when carbon monoxide and oxygen both are available to RBCs then the RBCs will bind 200 times faster with carbon monoxide than oxygen, thus forming carboxyhaemoglobin rather than oxyhaemoglobin. This carboxyhaemoglobin reduces the oxygen carrying capacity of RBCs (since RBCs contain haemoglobin, a red-coloured protein).

26. Correct option: (B) By burning petrol in automobiles

Leaded petrol used to be a source of high levels of lead in the air of major cities. Significant reductions in the levels of airborne lead were achieved over recent years as the result of air pollution legislation introduced to remove lead from petrol.

27. Correct option: (A) Increasing Forest reserves

Burning fossil fuels like coal release carbon dioxide and cause a large amount of air pollution. The construction of the housing estates releases harmful chemicals which result in air pollution. The quarrying of the limestone also results in the release of the particles causing air pollution. The increase in the forest reserves does not result in any kind of pollution. Rather planting more trees helps to control air pollution.

28. Incorrect options given

29. Correct option: (D) iii, i, ii

The amount of water vapor in the air is known as humidity. During condensation of the water molecules, water vapor combines to make tiny liquid water droplets that hang in the air. As we know water vapor is a gas and invisible but because of the tiny water droplets, we can see the fog with our eyes. Asthma is a chronic lung condition caused by Smog. Smog contains Smoke + Fog + Fumes.

30. Correct option: (C) iii, i, iv, ii

Carbon dioxide is greenhouse gas. CO2 is strongly absorbed by Infrared radiation. Nitrogen oxides affect vegetation. Sulphur dioxide is potent air pollutant.

8. Why do we fall ill?

Solutions

LEVEL-I

1. Correct option: (D) Elephantiasis

Lymphatic filariasis, commonly known as elephantiasis, is a painful and profoundly disfiguring disease. Elephantiasis can last for a very long time and has long-term effects on the body. Therefore, it is a chronic disease.

2. Correct option: (C) Jaundice

Jaundice happens when too much bilirubin is made in the blood or when the liver is damaged and cannot get rid of bilirubin from the body. Other than yellow skin, people with jaundice may have stomach pain, nausea, fever, weight loss, or itching.

3. Correct option: (A) Microbes

nfectious diseases are disorders caused by organisms — such as bacteria, viruses, fungi or parasites. Many organisms live in and on our bodies. They're normally harmless or even helpful. But under certain conditions, some organisms may cause disease.

4. Correct option: (C) Cancer

Mutations or genetic changes on DNA results in changing genetic makeup of a cell resulting in altering the behaviour and metabolism of cells which in return cause cells to divide indefinitely.

5. Correct option: (C) Robin Warren

Barry J. Marshall and Robin Warren, two Australian researchers who discovered the bacterium Helicobacter pylori and deciphered its role in gastritis and peptic ulcer disease, have been awarded this year's Nobel Prize in Physiology or Medicine.

6. Correct option: (B) 50

Upon analysis of negative-stained SARS-CoV-2 articles by electron microscopy, different researchers have had varying results, but the diameter of the virus has been found to range between 50 nm to 140 nm.

7. Correct option: (A) Small intestine

Roundworms are small organisms that can live in your intestine, which is part of your digestive system. Roundworms can live in the human small intestine for a long time and can cause infections. Ascaris is an intestinal parasite of humans. It is the most common human worm infection. The larvae and adult worms live in the small intestine and can cause intestinal disease.

8. Correct option: (A) bacteria

There are many organisms that can cause skin infections including bacteria, viruses and fungi. Bacteria like Staphylococcus aureus may cause skin infections, Fungi including Trichophytons cause ringworms, and viruses like Herpes cause cold sores. Most prominent infections are caused by bacteria.

9. Correct option: (B) tuberculosis

TB is spread through the air from one person to another. TB bacteria are passed through the air when someone who is sick with TB disease of the lungs or throat coughs, speaks, laughs, sings, or sneezes. Anyone near the sick person with TB disease can breathe TB bacteria into their lungs.

10. Correct option: (A) Mosquitoes

Most of the time mosquitoes feed on sugary substances like plant nectars. However, when it's time for female mosquitoes to lay eggs, they need the extra protein that comes from

blood supplies. Only female mosquitoes bite living hosts such as humans and other warm-blooded animals.

11. Correct option: (B) AIDS

HIV (human immunodeficiency virus) is a virus that attacks the body's immune system. If HIV is not treated, it can lead to AIDS (acquired immunodeficiency syndrome). There is currently no effective cure. Once people get HIV, they have it for life. But with proper medical care, HIV can be controlled.

12. Correct option: (A, B, D) Small pox, Chicken pox and AIDS

Small pox, Chicken pox and AIDS are viral diseases caused by Variola virus, varicella zoster virus and HIV respectively. Tuberculosis is a bacterial infection caused by Mycobacterium tuberculosis.

13. Correct option: (D) Diabetes

Diabetes is one of the four major types of noncommunicable diseases (cardiovascular disease, diabetes, cancer and chronic respiratory diseases). It is a chronic condition that occurs when the body either does not produce enough insulin or cannot effectively use the insulin it does produce.

14. Correct option: (D) Female Anopheles mosquito

In the case of malaria, the vector is the female anopheline mosquito and the disease-causing organism is the malaria parasite. Humans and anopheline mosquitoes are both considered to be the parasite's hosts.

15. Correct option: (C) TB

BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

16. Correct option: (A) Cholera

The cholera bacteria is passed through feces (poop). It is spread by eating or drinking food or water contaminated by the feces (poop) of an infected person. This occurs more often in underdeveloped countries lacking proper water supplies and sewage disposal.

17. Correct option: (D) Bed bugs

Plague is an infectious disease caused by the bacteria Yersinia pestis, a zoonotic bacterium, usually found in small mammals and their fleas. It is transmitted between animals through fleas. Humans can be infected through: the bite of infected vector fleas.

18. Correct option: (D) TB

Tuberculosis is a chronic, progressive mycobacterial infection, often with an asymptomatic latent period following initial infection. TB bacteria can live in the body without making you sick. This is called latent TB infection.

19. Correct option: (B) Alexander Flemming

In 1928 Dr Alexander Fleming returned from a holiday to find mould growing on a Petri dish of Staphylococcus bacteria. He noticed the mould seemed to be preventing the bacteria around it from growing. He soon identified that the mould produced a self-defence chemical that could kill bacteria and later on named it penicillin.

20. Correct option: (D) Cholera

Cholera is an acute diarrheal illness caused by infection of the intestine with Vibrio cholerae bacteria. People can get sick when they swallow food or water contaminated with cholera bacteria.

LEVEL-II

1. Correct option: (B) Increases

It can be hard to follow the work schedule, but the good thing is that you are at the comfort of your home, so utilize it at its best. Take a good 10-minute walk every 2 or 3 hours or do stretching and take a good diet and sanitary hygiene. This will not only increase your productivity but would also prevent you from diseases.

2. Correct option: (A) Good economic

Social harmony and good economic conditions are necessary for good health.

3. Correct option: (D) All the above

Community health is defined as the group of individuals living in a society who aims to maintain, protect and improve the health of the people. The whole community aims to maintain the health of people by providing safe drinking water, proper disposal of sewage, by providing proper medical facilities, and by providing proper health education to the people in the community.

4. Correct option: (C) Disturbed

The term "disease" is a condition in which the normal functioning of the body is disturbed. Many times, such conditions are not very serious and get cured, but sometimes they become serious and even fatal.

5. Correct option: (A) Short

Acute diseases come on rapidly, and are accompanied by distinct symptoms that require urgent or short-term care, and get better once they are treated.

6. Correct option: (D) Both A and B

Acute diseases last for a shorter period of time. They do not have any long-term effect on the patient. E.g., common cold, cough, etc.

7. Correct option: (B) Chronic

Reducing a person's ability to work efficiently and experiencing difficulties in learning at school are symptoms typically associated with chronic conditions or long-term issues rather than acute ones. They develop over time and can have a gradual onset. Symptoms like reduced work efficiency and learning difficulties can be linked to chronic conditions that affect a person's overall health, cognitive abilities, or emotional well-being over an extended period.

8. Correct option: (B) Chronic

Chronic diseases are typically long-lasting conditions that develop gradually and persist over time. Tuberculosis (TB) is caused by the bacterium Mycobacterium tuberculosis and primarily affects the lungs. It often presents with symptoms such as cough, weight loss, fatigue, and fever. If left untreated, TB can persist for an extended period, making it a classic example of a chronic disease.

9. Correct option: (B) Pathogens

The disease-causing organisms like bacteria, viruses, fungi, protozoans, worms, etc. are referred to as pathogens.

10. Correct option: (D) virus

Chickenpox is caused by the varicella-zoster virus (VZV). This virus is responsible for the development of chickenpox (varicella) during the initial infection and can later reactivate to cause shingles (herpes zoster) in some individuals.

11. Correct option: (B, D) Typhoid and Cholera

Cholera is caused by the bacterium Vibrio cholerae and primarily affects the digestive system, leading to severe diarrhea and dehydration. Typhoid fever is indeed caused by the

bacterium Salmonella typhi. It's a bacterial infection that can lead to high fever, abdominal pain, and other symptoms.

12. Correct option: (D) None of the above

Leprosy, Syphilis, and TB are all bacterial diseases caused by Mycobacterium leprae, Treponema pallidum and Mycobacterium tuberculosis respectively.

13. Correct option: (B) Unhealthy

A nutritious diet provides essential nutrients that the body needs to function properly, maintain good health, and prevent various health issues. When a person lacks proper nutrition due to an inadequate diet, their body may not receive the necessary vitamins, minerals, proteins, and other nutrients it requires.

14. Correct option: (A) Congenital

Congenital refers to a condition or trait that is present at or from birth. It can refer to both genetic conditions and conditions that might have been influenced by prenatal factors.

15. Correct option: (B) Acquired

Acquired diseases are those that are not present at birth and are typically developed or contracted due to various factors such as infections, lifestyle choices, environmental influences, and so on. These diseases are not inherited and do not pass from one generation to another through genetic means.

16. Correct option: (A) Communicable

A communicable disease (also known as an infectious or contagious disease) is a type of disease that can be transmitted from one person, animal, or object to another through various means. This can include direct contact, respiratory droplets, contaminated food or water, vectors like insects, and more. Some examples of communicable diseases include the common cold, influenza, tuberculosis, and COVID-19.

17. Correct option: (B) Infectious

Tuberculosis (TB) is an infectious disease that most often affects the lungs and is caused by a type of bacteria. It spreads through the air when infected people cough, sneeze or spit.

18. Correct option: (B) Non-infectious

Scurvy is a disease resulting from a lack of vitamin C (ascorbic acid). Early symptoms of deficiency include weakness, fatigue, and sore arms and legs. Thus, it is non-infectious.

19. Correct option: (C) Common cold

Colds are primarily transmitted from person to person via hands that are contaminated with nasal secretions. Less often, viruses that cause colds can be transmitted by touching a contaminated surface or via sneezing or coughing. Direct contact — People with colds typically carry the cold virus on their hands

20. Correct option: (C) Antibiotic

Penicillin is an antibiotic that is it is used to treat bacterial infections. It was discovered by Sir Alexander Fleming in the year 1920.

21. Correct option: (D) None

AIDS is a viral disease caused by HIV virus through blood transfusion via infected needles, sexual contact, and mother to infant through breast feeding. It is present in body fluids.

22. Correct option: (A) Water borne

Drinking water should be filtered and boiled, ozonised, and treated with chlorine before making it potable, thereby avoiding water borne infections. This ensures that no pathogenic bacteria like typhus and cholera are persisting there.

23. Correct option: (A) transmissible

Communicable diseases, also known as infectious diseases or transmissible diseases, are illnesses that result from the infection, presence and growth of pathogenic (capable of causing disease) biologic agents in an individual human or other animal host.

24. Correct option: (C) Are vectors for many diseases

Mosquitoes are known to transmit various diseases such as malaria, dengue fever, Zika virus, and more. Preventing mosquito breeding in our surroundings is crucial to reduce the spread of these diseases.

25. Correct option: (B) Tuberculosis

BCG, or bacille Calmette-Guerin, is a vaccine for tuberculosis (TB) disease. Many foreign-born persons have been BCG-vaccinated. BCG is used in many countries with a high prevalence of TB to prevent childhood tuberculous meningitis and miliary disease.

9. Improvement in Foor Resources

Solutions

LEVEL-I

1. Correct option: (D) Phosphorous

Phosphorus is one of the three macronutrients essential for plant growth. It is required for the photosynthesis process, converting the sun's energy into food for the plant. It is also required for strong root development. A plant must be able to access phosphorus to ensure a healthy growing cycle.

2. Correct option: (D) Phosphorous

Phosphorous is used by plants to help form new roots, make seeds, fruit and flowers. It's also used by plants to help fight disease.

3. Correct option: (B) Black Gram

Black gram can be grown on variety of soils ranging from sandy soils to heavy cotton soils. The most ideal soil is a well-drained loam with pH of 6.5 to 7.8. Black gram cannot be grown on alkaline and saline soils. Land is prepared like any other kharif season pulse crop.

4. Correct option: (D) All of these

Intercropping is the practice of growing two or more crops in proximity. For instance, the spaces between the rows of the Jackfruit trees should be cultivated and used for the production of annual or perennial intercrops or both. Examples of such intercrops are Citrus, Banana, Pineapple, Corn, Peanut (Groundnut), and other Pulses, Spices, and vegetables.

5. Correct option: (D) Both A & B

Root nodules are mostly present in the leguminous plants. E.g. Pulses, beans, soybeans, etc. These nodules provide shelter to certain bacteria which aids in nitrogen fixation.

6. Correct option: (A) Rhizobia

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

7. Correct option: (D) Both A & B

Manure is an organic substance obtained by the decomposition of animal and plant wastes. It helps to maintain the fertility of the soil without having deleterious effects on soil as well as the crops grown on it.

8. Correct option: (D) All of these

These concentrated manures are produced using basic materials that are either plant- or animal-derived. Oil cakes, bone meal, fish meal, blood meal, and meat meal are some of the concentrated organic manures that are frequently employed. These include Jatropha, Neem and Groundnuts.

9. Correct option: (D) Both A and C

Natural organic manures are generally divided into two types. One is concentrated organic manures and the other is macro-organic manure. Animal excreta, compost, deep litter are the examples of macro-organic manure.

10. Correct option: (C) Insecticides

Insecticides are pesticides that are formulated to kill, harm, repel or mitigate one or more species of insect. Insecticides work in different ways. Some insecticides disrupt the nervous

system, whereas others may damage their exoskeletons, repel them or control them by some other means.

11. Correct option: (D) All of these

The three options that have been used for increasing food production are (i) agro-chemical based agriculture (ii) organic agriculture; and (iii) genetically engineered crop-based agriculture.

12. Correct option: (B) Carbohydrate

Starch is a carbohydrate commonly found in nature and one of the primary sources of food energy for human beings. It is regularly eaten in the form of wheat, rice, potatoes, and other staple foods cultivated throughout the world. Alongside fiber and sugar, starch is one of the three main categories of carbohydrates.

13. Correct option: (C) Drip irrigation

Drip irrigation saves wastage of water because it supplies water directly to the roots.

14. Correct option: (B) Nitrogen

The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth.

15. Correct option: (D) All of these

Fertilisers and manures contain plant nutrients such as nitrogen, phosphorous and potassium, etc. So, when fertilisers and manures are added to the soil in the fields, then the soil gets enriched with nutrients like nitrogen, phosphorous and potassium, etc.

Nutrients are replenished by certain different microorganisms that are found in the soil or also by the death of certain bacteria such as the rhizobium.

16. Correct option: (B) 50 to 150 kg

Values estimated for various legume crops and pasture species are often impressive, commonly falling in the range of 50 to 150 kg of N ha^{-1} $year^{-1}$.

17. Correct option: (D) All the above

Panchagavya or panchakavyam is a mixture used in traditional Hindu rituals that is prepared by mixing five ingredients. The three direct constituents are cow dung, urine, and milk; the two derived products are curd and ghee. These are mixed in proper ratio and then allowed to ferment.

18. Correct option: (B) 46%

The percentage of nitrogen present in urea is 46.6%. Urea contains four elements: carbon, hydrogen, nitrogen, and oxygen. The molecular formula for urea is CH4 N2 O. This compound is made up of a carbonyl (C=O) group attached to two nitrogen groups (NH2). A carbonyl carbon is one that is double bonded to oxygen.

19. Correct option: (A) Tungrovirus on paddy

Tungro virus disease is transmitted by leafhoppers, wherein the most efficient vector is the green leafhopper.

20. Correct option: (A) Rhizobium

Rhizobium is a soil bacteria that fixes nitrogen belonging to the genus of gram-negative bacteria. Rhizobium species from endosymbiotic nitrogen-fixing associations with legumes. The bacteria colonize plant cells in the roots, where they use the nitrogenase enzyme to convert atmospheric nitrogen to ammonia and then provide organic nitrogen compounds such as glutamine.

21. Correct Matching is: a-T, b-S, c-P, d-R, e-Q

(a) Kharif crop – (T) Agricultural crops that arise in the rainy season

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).

(b) Rabi crop – (S) Agricultural crops that raise in winter

Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May

(c) Mixed crop – (P) Cultivation of more than one crop in the same field

In agriculture, multiple cropping or multicropping is the practice of growing two or more crops in the same piece of land during one year, instead of just one crop. When multiple crops are grown simultaneously, this is also known as intercropping.

(d) Weedicides – (R) Chemicals used to kill weeds

Weedicides are the chemicals which are generally sprayed all around the field to prevent the growth of weeds. An example of common weedicide is 2,4- D (2,4 dichlorophenoxy acetic acid) which acts as a dicot weedicide.

(e) Herbicides – (Q) Chemicals used to kill herbs

Herbicide stops the growth of plants whereas weedicides skill weeds. They are designed to kill specific types of pests and are used mainly on plants.

22. Correct option: a-S, b-P, c-T, d-Q, e-R

(a) Earthworm – (S) Vermicompost

Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.

(b) Urine of cow – (P) Panchagavya

It consists of nine products viz. cow dung, cow urine, milk, curd, jaggery, ghee, banana, Tender coconut and water. When suitably mixed and used, these have miraculous effects.

(c) Gingelly – (T) Concentrated organic manure

(d) Zinc – (Q) Micronutrient

Zinc (Zn) is an important micronutrient for plants since it is involved in many key cellular functions such as metabolic and physiological processes, enzyme activation, and ion homeostasis.

(e) Phosphorous – (R) Macronutrient

The primary macronutrients are Nitrogen (N), Phosphorus (P), and Potassium (K). Nitrogen is essential for plant development, since it plays a fundamental role in energy metabolism and protein synthesis. Nitrogen is absorbed by the plant in the form of a nitrate. This macronutrient is directly related to plant growth

LEVEL-II

1. Correct option: (C) Both A and B

Rice has approximately 2.69% of protein, and they have high carbohydrates. At the same time, millets are low in carbohydrates than rice and contain a higher amount of protein.

2. Correct option: (B) Green gram

The mung bean or Organic green gram is a legume family plant. Its super high protein content has made it a popular ingredient for making plant-based alternatives to animal-based food items like eggs.

3. Correct option: (A) Rainy

In India, the season is popularly considered to start in June and end in October. Kharif crops are usually sown at the beginning of the first rains during the advent of the south-west monsoon season, and they are harvested at the end of monsoon season (October–November).

4. Correct option: (C) Winter

Rabi Crops are harvested in the spring season while it is sown in winter. The rabi crops are sown around mid-November, preferably after the monsoon rains are over, and harvesting begins in April / May.

5. Correct option: (C) Peas

Examples of Kharif crops

Paddy, maize, bajra, jowar, soybean, castor, cotton, sugarcane, turmeric, chilly, bitter gourd, guar, okra, brinjal,turmeric, ragi.

6. Correct option: (B) Hybridization

Hybridization refers to crossing between genetically dissimilar plants. This crossing may be intervarietal (between different varieties), interspecific (between two different species of the same genus) or intergeneric (between different genera).

7. Correct option: D) All the above

Cultivation practices and crop yield are related to weather, soil quality and availability of water. Since weather conditions such as drought and flood situations are unpredictable, varieties that can be grown in diverse climatic conditions are useful.

8. Correct option: (C) Both A and B

Crop management is the set of agricultural practices performed to improve the growth, development and yield of crops. It begins with a seedbed preparation, sowing of seeds and crop maintenance; and ends with crop harvest, storage and marketing.

9. Correct option: (B) Manure

Manure is obtained when the animal and plant waste are dumped by farmers in an open pit and leave it for days. The waste starts to decompose by the action of certain microorganisms, this process is named decomposition. Finally, the decomposed organic matter becomes Manure.

10. Correct option: (D) Both A and B

Air supplies two nutrients to plants namely carbon and oxygen. Water supplies hydrogen and soil supplies thirteen nutrient elements to the plants. Among these thirteen nutrients, six are macronutrients and the other seven nutrients are micronutrients.

11. Correct option: (B) Soil

Plants derive the required nutrients from the air, water, and soil. There are sixteen nutrients essential for the growth of plants. Carbon and Oxygen are supplied by water. The remaining thirteen nutrients are supplied by soil.

12. Correct option: (A) Macronutrients

Nutrients which are needed in large quantities are called macro nutrients.

13. Correct option: (B) Calcium

These are the most important elements required for crops and examples include nitrogen (N), potassium (K), calcium (Ca), phosphorus (P), magnesium (Mg), sulphur (S), oxygen (O), carbon (C), and hydrogen (H). Calcium is an essential micronutrient in plants. It serves as a constituent of cell walls and membranes and thus contributes to the structure of cells and the upholding of physical barriers against pathogens.

14. Correct option: (B) Copper

Copper (Cu) is one of eight essential plant micronutrients. Copper is required for many enzymatic activities in plants and for chlorophyll and seed production. Deficiency of copper

can lead to increased susceptibility to diseases like ergot, which can cause significant yield loss in small grains.

15. Correct option: (B) Inorganic

It is commonly prepared by decomposing plant and food waste, recycling organic materials, and manure. The resulting mixture is rich in plant nutrients and beneficial organisms, such as bacteria, protozoa, nematodes, and fungi.

16. Correct option: (A) Vermicompost

Vermicomposting is a natural process whereby earthworms convert waste material with rigid structures into compost. The compost produced in this green process is traditionally and popularly used as a natural fertilizer for enhancing plant growth.

17. Correct option: (C) Fertilizers

Fertilizers supply nitrogen, phosphorus and potassium. They are used to ensure good vegetative growth (leaves, branches and flowers), giving rise to healthy plants. Fertilizers are a factor in the higher yields of high-cost farming.

18. Correct option: (D) All the above

Excess use of fertilizers causes environmental pollution as their residual and unused amounts will become pollutants for air, water, and soil. It also initiates the process of eutrophication in the water bodies, decreasing the soil quality by killing the soil microflora, etc.

19. Correct option: (C) 3

There are three types of wells: dug, driven, and drilled. Drilled are the most common today.

20. Correct option: (A) Mixed cropping

Growing two or more crops simultaneously on the same piece of land is known as mixed cropping. This lowers the risk and provides some protection if one of the crops fails. For example - Wheat with gram.

21. Correct option: (B) Inter- Cropping

The practice of growing two or more crops simultaneously in definite rows in the same field is called inter cropping.

22. Correct option: (B) Crop rotation

Growing different crops on a piece of land in pre-planned succession is known as crop rotation.

23. Correct option: (D) All of them

Weeds are unwanted plants which grow along with the crop plants in the cultivated field. Some common weeds in India are Amaranthus (Chaulai), Parthenium (Gajar ghas), Cyperus rotundus (Motha), Chenopodium (Bathua), Xanthium (Gokhroo). Weeds compete with the crop plants for nutrients, space, light and water.

24. Correct option: (A) Animal

Animal husbandry is a scientific management of livestock, breeding and rearing of farm animals.

25. Correct option: (D) Both A and C

Most of the Indian beekeepers engaged in beekeeping activity are primarily doing the extraction of raw honey from the bee colonies maintained by them. Very few beekeepers are extracting wax & pollen.